Computational modeling of neural populations

Lecture notes

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Contents

1 Context **5**

1.1 Model categorization 5

1.2 Mathematical formalisms in computational modeling 6

1.3 Rationale for computational approaches 6

1.4 Multiscale perspectives: micro, meso and macro 7

1.5 Approaches to modeling neural populations 8

1.6 Practical applications and course roadmap 9

2 Action potential and synaptic interactions **10**

2.1 Membrane potential and ion gradients 11

2.2 The action potential 13

2.3 Refractory periods 13

2.4 Chemical synaptic transmission 14

2.5 Excitatory and inhibitory postsynaptic potentials 16

2.6 Postsynaptic integration 17

2.7 From single neurons to population models 17

3 Localized Wilson-Cowan model **19**

3.1 Modeling assumptions 20

3.2 Mathematical building blocks 20

3.2.1 Refractoriness 21

3.2.2 Population transfer functions 22

3.2.3 From synaptic input to population activity 24

3.3 Time coarse-graining . . . 26
3.3.1 The refractory and synaptic time windows . . . 26
3.3.2 A differential equation for the averaged activity . . . 27
3.4 Reading the localized equations . . . 28
3.5 Temporal activity patterns . . . 28
3.6 Scope of the localized model . . . 29

4 Spatially extended Wilson-Cowan model **30**
4.1 Why add space? . . . 30
4.2 Modeling assumptions and activity variables . . . 30
4.3 Distance-dependent neural connectivity . . . 31
4.4 From temporal history to field equations . . . 32
4.5 Reading the spatial equations . . . 34
4.6 Spatiotemporal activity patterns . . . 35
4.7 Scope of the spatial model . . . 35

5 Amari model **36**
5.1 From two populations to a single population . . . 36
5.2 Modeling assumptions and field equation . . . 36
5.3 Population transfer function and neural connectivity . . . 37
5.4 Equilibrium solutions . . . 38
5.4.1 The kernel integral . . . 40
5.4.2 Existence statements and their reasoning . . . 40
5.5 Stability of equilibrium solutions . . . 43
5.6 Scope of the model . . . 49

6 Neurogeometry **50**
6.1 From the retina to orientation-selective cells . . . 50
6.2 Receptive fields, profiles, and orientation selectivity . . . 51
6.3 Gabor function models a simple-cell receptive profile . . . 52
6.4 Orientation columns and orientation fibres . . . 53

6.5 Lifting an image into position-orientation space 54

Chapter 1

Context

1.1 Model categorization

In neuroscience, modeling is not merely a descriptive convenience but a strategic necessity. The complexity of the human brain demands a rigorous approach to abstraction, where the primary challenge lies in selecting the appropriate model type to address a specific research question. The decision to employ biological versus computational frameworks involves a fundamental trade-off: anchoring a study in biological ground truth or pursuing the theoretical clarity afforded by mathematical formalization.

Biological models serve as the empirical foundation of our understanding. They provide ground truth through physical interventions across various levels: *in-vivo* (animal models), *ex-vivo* (tissue preparations), and *in-vitro* (cell cultures). By utilizing specific model organisms such as flies, worms, mice, rats, and monkeys, researchers can observe biological mechanisms in their native contexts. However, such biological preparations are limited by physical constraints.

Computational models provide a vital framework for abstraction. Critically, computational approaches allow for the exhaustive exploration of parameter spaces that would be physically impossible, or ethically prohibitive, to investigate in biological systems. This makes computational modeling inevitable for a complete understanding of mechanisms underlying biological phenomena.

We distinguish between two primary computational modeling approaches: phenomenological and biophysical. Phenomenological models, often referred to as toy or minimal models, prioritize the representation of observed behavior over structural/anatomical precision. On the contrary, biophysical models attempt to replicate the exact physiological mechanisms of the neural system. See Table 1.1.

As we move from these conceptual categorizations toward practical implementation, it is imperative to formalize these abstractions by using the precise mathematical languages that describe neural activity.

Table 1.1: **Phenomenological and biophysical modeling.**

Model type	Phenomenological	Biophysical
Complexity	Simple	Complex
Representation	Representative of patterns, behavior	Focused on exact mechanisms
Precision	Approximate	(More) Precise
Detail	Minimum detail	Highly detailed

1.2 Mathematical formalisms in computational modeling

The temporal and spatial evolution of neural dynamics is governed by differential equations, which are described by (possibly) nonlinear functions, one of which we will denote by f. We adopt specific formalisms to represent neural dynamics:

- ODEs (Ordinary Differential Equations): These govern temporal dynamics, represented as
$$\frac{\mathrm{d}u}{\mathrm{d}t} = f(u, t),$$
where the variable $u(t)$ tracks neural state changes strictly over time (t). There is no spatial component.
- PDEs (Partial Differential Equations): These are required for spatio-temporal dynamics, represented as
$$\frac{\partial u}{\partial t} = f(u, x, t),$$
where $u(x, t)$ accounts for changes across both space (x) and time.
- IDEs (Integro-Differential Equations): These formalize nonlocal interactions by incorporating an integral kernel to account for spatial influences over a distance:
$$\frac{\partial u}{\partial t} = f(u, x, t) + \int [\dots] \, \mathrm{d}y.$$
Amari and Wilson-Cowan models are famous examples of IDEs.
- SDEs (Stochastic Differential Equations): These are ODEs, PDEs, or IDEs supplemented with noise terms to represent biological variability, or to study noise-related phenomena such as neural excitability.

These mathematical frameworks provide the necessary foundation for the iterative scientific process that links theory to observation.

1.3 Rationale for computational approaches

In modern theoretical biology, a deep symbiotic relationship exists between experimental data and computational modeling. Neither can fully explain brain function in isolation. Instead, they function within an iterative modeling loop that drives the evolution of our hypotheses.

This cycle, as visualized in Figure 1.1, begins with experimental data which is derived from in-vivo, in-vitro, or clinical sources, and which informs a hypothesis. This leads to model design, followed by model simulation. Simulations are essential for model analysis and model fitting to the experimental data. Resulting predictions must then undergo experimental validation. This ultimately leads back to model refinement. All this procedure allows us to narrow the search space for discovery, before returning to the laboratory.

The strategic rationale for employing this computational loop can be distilled into four primary objectives:

1. Data synthesis: achieving an understanding and organization of complex, high-dimensional datasets,
2. In-silico exploration: studying what-if scenarios that are physically or ethically impossible in biological experiments,
3. Prediction: generating precise predictions through the absolute control of parameters,
4. Experimental optimization: facilitating and guiding future biological experiments by narrowing the scope of potential mechanisms.

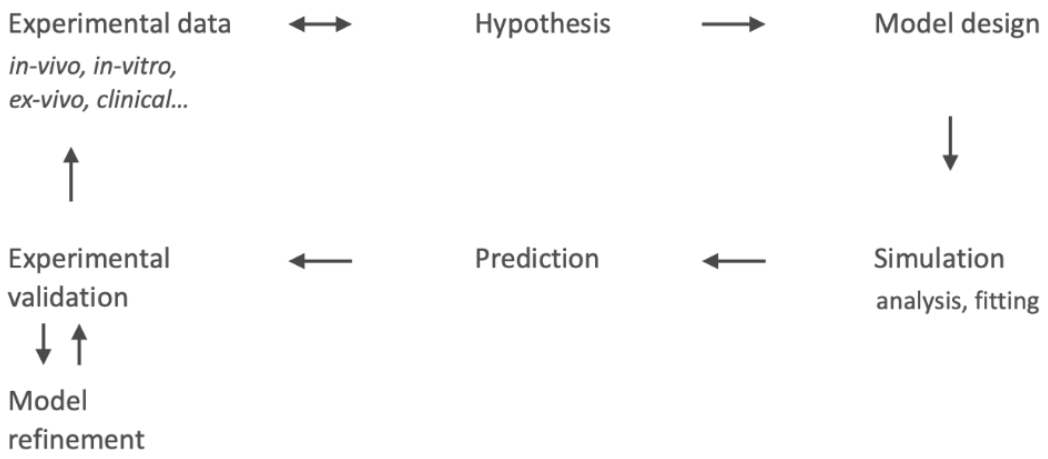


Figure 1.1: **Modeling cycle.**

These tools offer a robust predictive framework. Nevertheless, their utility depends on the physical scale at which the neural mechanisms are being investigated.

1.4 Multiscale perspectives: micro, meso and macro

Scale is a fundamental concept in neuroscience. All biological phenomena in neuroscience have a multiscale nature. Bridging observations made at different scales remains the central challenge of the field. As a result, the choice of modeling strategy differs radically in the range from micro- to macro-scale; see Table 1.2.

The three primary scales of observation involve distinct entities and recording signals:

Table 1.2: **Multiple scales in neuroscience.**

Scale	**Target**	**Signal type**
Micro-scale	Single neuron	Action potential, membrane potential
Meso-scale	Neural populations	Local field potential (LFP), EEG, multielectrode array
Macro-scale	Whole brain	MRI, fMRI, PET

The meso-scale is particularly critical. It serves as the bridge where individual neurons transition into functional signals such as LFP and EEG. It is at this level that we attempt to capture emergent properties or functional dynamics of the neural population that are entirely invisible when observing individual neurons at the micro-scale. Our focus will be on meso-scale models throughout the course.

1.5 Approaches to modeling neural populations

When modeling groups of neurons, we face a trade-off between biophysical realism and computational tractability. Two primary paradigms dominate the literature (Figure 1.2):

- Networks: Networks are high-dimensional systems that track the state of every individual neuron represented as a system of differential equations. They provide biophysically detailed representations and precise dynamics. However, they are computationally heavy and difficult to manipulate.
- Low-dimensional approximations: These models focus on the collective, averaged behavior of a population. They are low-dimensional, analytically simple, and computationally light. However, they lack individual biophysical detail and provide only approximate dynamics.

The application of averaged dynamics is a powerful tool for dimensionality reduction. By averaging population activity, we maintain the representative behavior of the system while grouping micro-scale details in a phenomenological parameter.

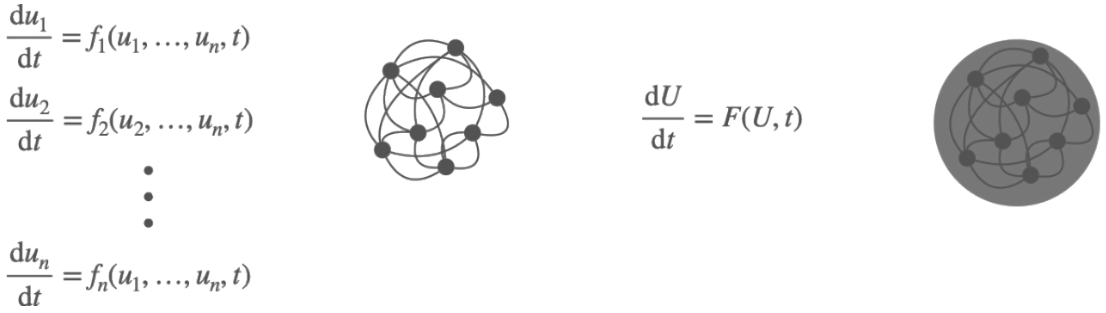


Figure 1.2: **Schematization of a network and a low-dimensional approximation.** In networks, each neuron is represented by a separate variable: u_1, u_2, $\ldots$, u_n. In low-dimensional approximations, average activity U of the neurons is described at a coarse-grained continuum limit.

1.6 Practical applications and course roadmap

Computational population models are not mere abstractions. They translate directly into clinical and cognitive insights. By using low-dimensional approximations, we can effectively model several complex phenomena:

- Neural activity patterns: periodic spiking and the movement of propagating waves across the cortex.
- Visual perception: orientation detection and spatial processing within the visual cortex.
- Brain states: mechanisms governing the transitions between awake states and sleep or anesthesia.

The structure of this course is as follows:

- Part I. Neural fields: We will examine the Wilson-Cowan model, the spatially extended Wilson-Cowan model, and the Amari model to understand neural population dynamics.
- Part II. Neurogeometry: We will introduce the neurogeometric approach to study orientation detection in the visual system.
- Part III. Whole-brain dynamics: We will get acquainted with The Virtual Brain (TVB) as an example tool to model brain states.

Chapter 2

Action potential and synaptic interactions

Neural population models describe the activity of many neurons by using a small number of variables. To understand these variables, we first need to follow the signal produced by one neuron. A neuron receives synaptic inputs, combines them, and consequently may generate an action potential. The action potential then travels along the axon and can influence other neurons through synapses.

This chapter introduces the physiological ideas that will be used in the Wilson-Cowan and Amari models. We focus on four points: the membrane potential, the action potential, the refractory period, and the integration of excitatory and inhibitory synaptic inputs.

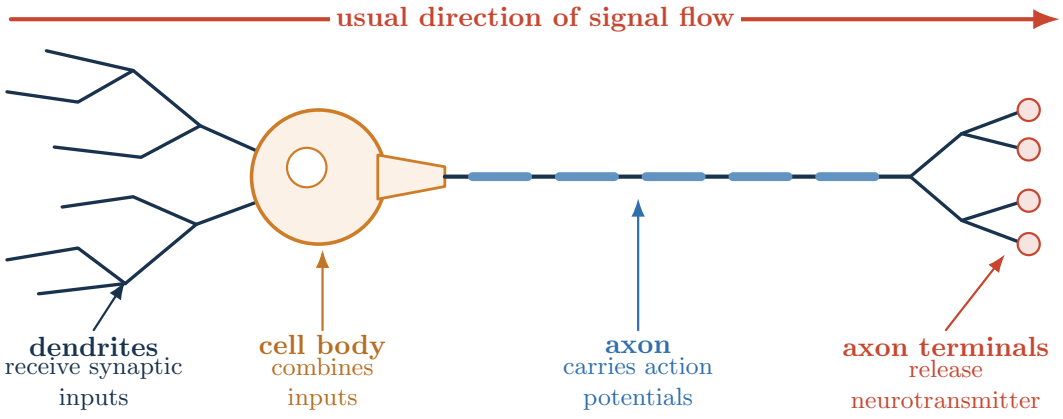


Figure 2.1: **Signal flow through a neuron.** Dendrites receive synaptic inputs. The cell body combines these inputs, the axon carries action potentials, and the axon terminals communicate with target cells.

2.1 Membrane potential and ion gradients

The membrane of a neuron separates the fluid inside the cell from the fluid outside it. The two sides contain different concentrations of ions. In a simplified description, the concentration of sodium ions (Na^+) is higher outside the neuron, whereas the concentration of potassium ions (K^+) is higher inside it. The membrane also contains ion channels that can allow selected ions to pass.

The membrane potential (V_m) is the electrical potential inside the neuron relative to the outside:

$$V_m = V_{\text{in}} - V_{\text{out}},$$

where V_{in} and V_{out} denote the electrical potential inside and outside the neuron, respectively. For a neuron at rest, a typical value of V_m is about -70 mV. Thus, the inside is electrically negative relative to the outside. The resting value results mainly from the unequal ion concentrations and from the selective permeability of the membrane. The sodium-potassium pump helps maintain the concentration gradients, and thus the resting membrane potential, by moving three Na^+ ions out of the cell and two K^+ ions into it for each cycle. Since it moves ions from low to high concentration areas, it uses cellular energy (ATP).

Some ion channels open and close in response to changes in membrane potential. These are called *voltage-gated ion channels*. Their opening and closing produce the rapid, further change of membrane potential that we call an *action potential*; see Figure 2.3.

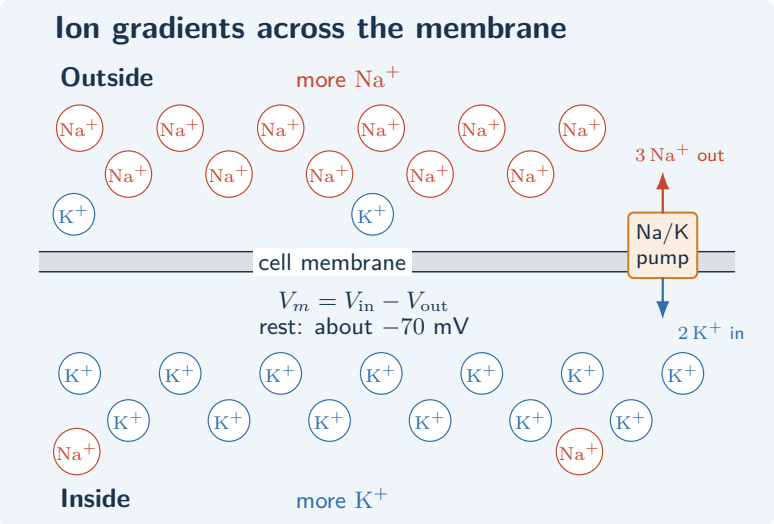


Figure 2.2: **Ion gradients and the resting membrane potential.** Sodium ions are more concentrated outside the cell, while potassium ions are more concentrated inside. The sodium-potassium pump maintains these concentration gradients and contributes to the resting membrane potential.

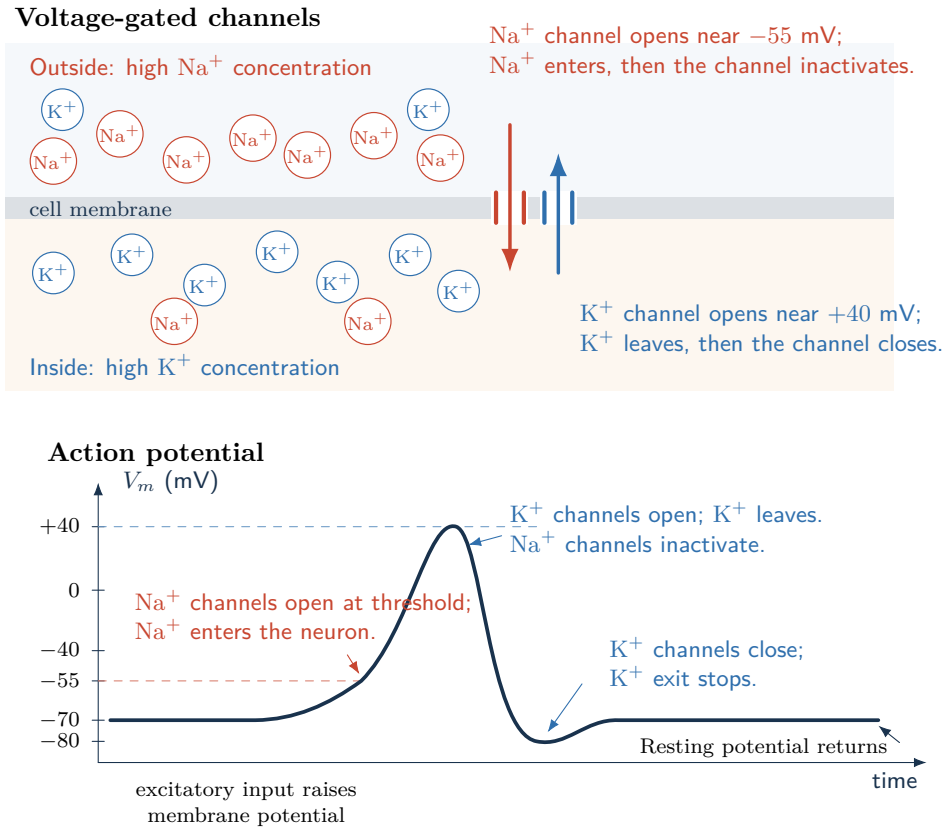


Figure 2.3: **Voltage-gated channels and the action potential.** Sodium entry drives the rise in membrane potential; potassium exit drives its fall.

2.2 The action potential

An excitatory synaptic input can move the membrane potential toward less negative values. This change is called *depolarization*. If the voltage at the axon initial segment reaches a threshold, typically near -55 mV, voltage-gated sodium channels open. Sodium ions enter the cell because of both their concentration gradient and the electrical force across the membrane.

The action potential develops in a short sequence:

1. **Rest.** The membrane potential is close to -70 mV.
2. **Threshold crossing.** Sufficient depolarization increases the membrane potential to the threshold -55 mV. Once the threshold is reached, voltage-gated Na^+ channels open and allow Na^+ to enter the cell.
3. **Depolarization.** The entry of Na^+ produces a rapid rise of the membrane potential, often to about $+40$ mV.
4. **Repolarization.** As the membrane potential approaches the peak value $+40$ mV, sodium channels become inactivated and voltage-gated K^+ channels open. K^+ leaves the cell, so the membrane potential falls towards the resting value.
5. **Hyperpolarization and return to rest.** K^+ channels close gradually. The potential may briefly become more negative than its resting value before returning to rest. This results in a brief undershoot, which is called *hyperpolarization*. The membrane potential comes eventually back to the resting value -70 mV after hyperpolarization.

Once threshold has been crossed, the action potential has an approximately fixed size. A stronger input does not normally make one action potential much larger. Instead, it can make the neuron fire sooner or produce more action potentials over time. This is called the *all-or-none* property.

The depolarized region of an axon generates local electrical currents that depolarize the next region. In this way, the action potential is regenerated along the axon and reaches the axon terminals without gradually fading away. This chain effect of depolarization on the axon is called *electrotonic conduction*.

2.3 Refractory periods

Immediately after an action potential, the neuron cannot generate another one in the same way. During the *absolute refractory period*, most voltage-gated Na^+ channels are inactivated. A second action potential cannot occur, even if the excitatory input is strong. During the following *relative refractory period*, some Na^+ channels have recovered, but the membrane is often still more negative than its resting value. A new action potential is possible, but it requires a stronger input than usual; see Figure 2.4.

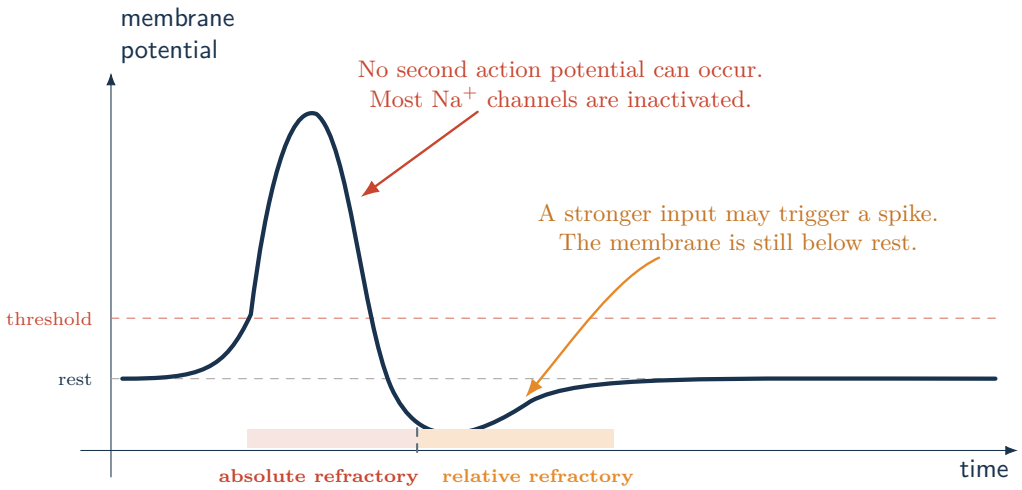


Figure 2.4: **Absolute and relative refractory periods.** Na^+-channel inactivation prevents a second spike during the absolute refractory period. During the relative refractory period, a sufficiently strong excitatory input may generate a new spike.

Refractoriness has two important consequences. First, it helps action potentials move in one direction along the axon because the region that has just fired is temporarily unable to fire again. Second, it limits how rapidly a neuron can produce successive action potentials. **In a population model, refractoriness means that only a fraction of the neurons is available to fire at a given time.**

2.4 Chemical synaptic transmission

Neurons usually communicate at specialized junctions called *synapses*. At a chemical synapse, the neuron sending the signal is the *presynaptic neuron*. The neuron receiving the signal is the *postsynaptic neuron*. The small space between their membranes is the *synaptic cleft*; see Figure 2.5.

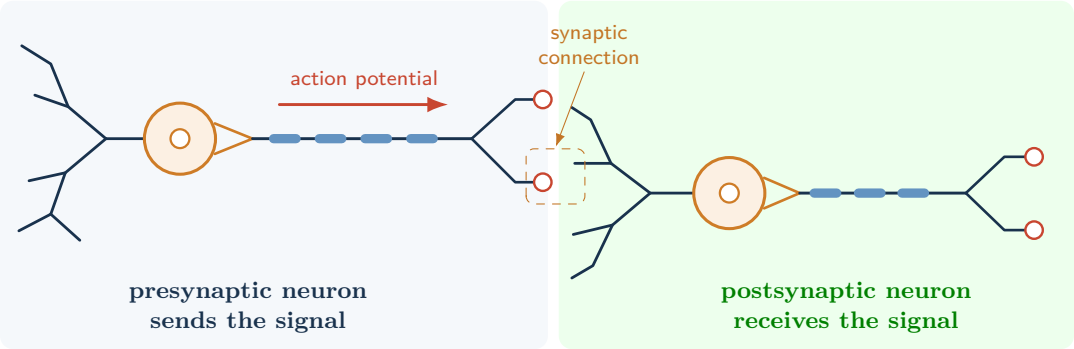


Figure 2.5: **Chemical synapse between two neurons.** The presynaptic neuron sends an action potential toward its axon terminals. The axon terminals face dendritic branches of the postsynaptic neuron.

An action potential travels along the presynaptic axon to axon terminals facing dendrite branches of the postsynaptic neuron. At the junction between presynaptic and postsynaptic neurons, each presynaptic axon terminal contains small *synaptic vesicles* filled with neurotransmitter molecules. Voltage-gated Ca^{2+} channels lie in the membrane of the axon terminal. Across the synaptic cleft, the postsynaptic membrane has receptors that recognize the released neurotransmitter molecules from vesicles, as shown in Figure 2.6.

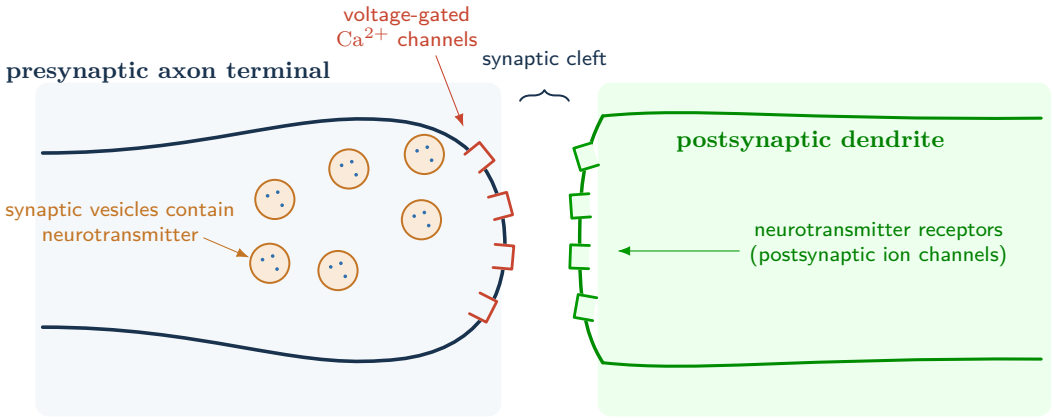


Figure 2.6: **Anatomy of the synapse.** Vesicles store neurotransmitter in the axon terminal. The presynaptic membrane has calcium channels, and the postsynaptic membrane has neurotransmitter receptors. A narrow synaptic cleft separates the membranes.

Synaptic transmission of a signal from a presynaptic to a postsynaptic neuron follows the sequence shown in Figure 2.7:

1. An action potential reaches the presynaptic axon terminal and depolarizes its membrane.
2. Voltage-gated Ca^{2+} channels open.
3. Ca^{2+} ions enter the presynaptic terminal.
4. This Ca^{2+} influx causes synaptic vesicles to fuse with the presynaptic membrane.

5. The vesicles release neurotransmitter molecules into the synaptic cleft.
6. The neurotransmitter molecules diffuse across the cleft and binds to receptors on the postsynaptic membrane.
7. The receptors change the opening of postsynaptic ion channels. This produces a local change of membrane potential.

The same process can occur at many synapses on the postsynaptic neuron. The resulting local changes of postsynaptic membrane potential combine. If their combined effect reaches the firing threshold, the postsynaptic neuron generates an action potential, which can carry the signal to other neurons through its own synapses.

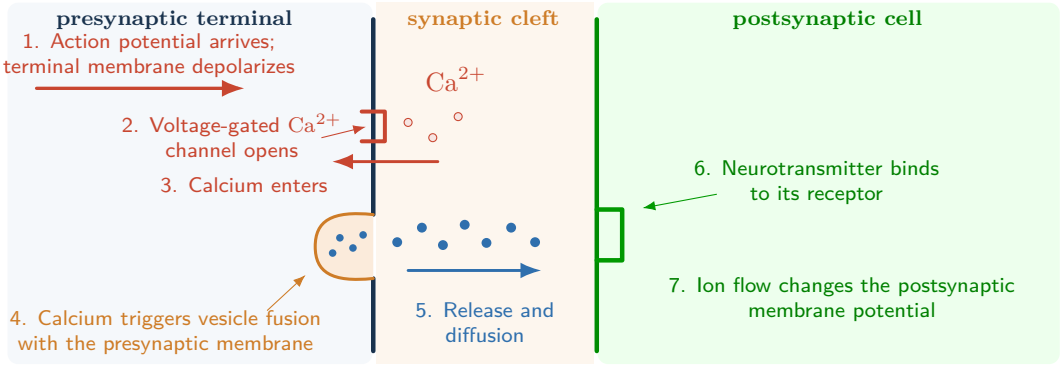


Figure 2.7: **Transmission across the synaptic cleft.** The numbered labels match the sequence in the text. A fused vesicle releases neurotransmitter into the cleft; receptor activation changes the postsynaptic membrane potential.

The neurotransmitter signal must also end. Transporters can take neurotransmitter molecules back into cells, enzymes can break them down, or they can diffuse away from the cleft. These mechanisms allow the synapse to respond to later action potentials.

2.5 Excitatory and inhibitory postsynaptic potentials

The local voltage change produced at a synapse is called a *postsynaptic potential* (PSP). Its magnitude depends on the synaptic input, and it decreases with increasing time and distance.

An *excitatory postsynaptic potential* (EPSP) moves the membrane potential toward the firing threshold, i.e. increases the membrane potential towards less negative values. Postsynaptic excitatory receptors often open channels that allow positively charged ions to enter the postsynaptic neuron. The postsynaptic neuron then becomes more likely to generate an action potential. This justifies the term “excitatory”.

An *inhibitory postsynaptic potential* (IPSP) moves the membrane potential away from threshold, i.e. decreases the membrane potential towards more negative values. This makes further depolarization more difficult. Depending on the postsynaptic receptor type and ion concentrations, inhibition commonly involves chloride (Cl^-) entry or potassium (K^+) exit. The postsynaptic neuron then becomes less likely to fire. This justifies the term “inhibitory”.

The words *excitatory* and *inhibitory* describe the effect on the probability of firing. They do not refer only to the direction in which one type of ion moves.

2.6 Postsynaptic integration

A postsynaptic neuron receives many EPSPs and IPSPs through its dendrites and cell body. These PSPs combine as they spread toward the initial segment of the axon of the postsynaptic neuron. Inputs arriving at different synapses combine through **spatial** summation. Repeated inputs arriving close together in time combine through **temporal** summation.

The resulting postsynaptic membrane potential after these spatial and temporal summations reflects the balance between excitation and inhibition. If the combined input crosses the firing threshold at the initial segment of the axon, the neuron generates an action potential. If it remains below threshold, no action potential is produced. Figure 2.8 illustrates this conversion from many synaptic potentials to one all-or-none output spike.

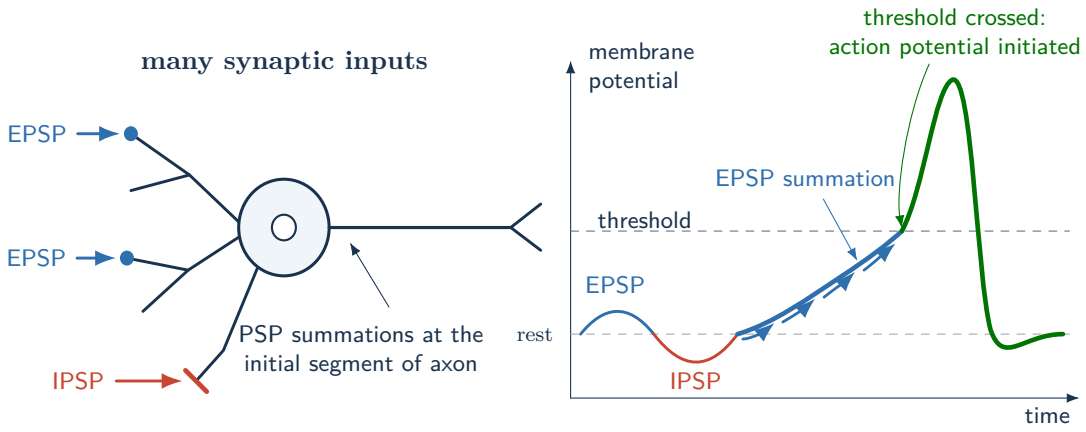


Figure 2.8: **Postsynaptic integration.** EPSPs and IPSPs combine in the receiving (postsynaptic) neuron. If the resting membrane potential increases enough after the spatial and temporal summations of EPSPs and IPSPs, it crosses the threshold at the axon initial segment. In this case the neuron generates an action potential.

2.7 From single neurons to population models

The preceding mechanisms motivate the main ingredients of a neural population model:

- **Excitatory and inhibitory input.** EPSPs increase the tendency to fire, while IPSPs decrease it. At the population level, their average effects enter with opposite signs.
- **A firing threshold.** A neuron produces an action potential only when its membrane potential reaches threshold after PSP summation. In a population, neurons have different thresholds and receive different inputs, so the response changes gradually as the mean input increases.

- **Refractoriness.** A neuron that has just fired is temporarily unavailable to fire again. Therefore, the number of neurons that can respond to an input depends on the recent firing history of the population.
- **Synaptic time course.** A postsynaptic potential lasts for a finite time. The present response therefore depends on recent inputs, not only on the input arriving at the present instant.

These ideas will reappear in Chapter 3. The Wilson-Cowan model represents the average activities of excitatory and inhibitory populations. A population transfer function converts the mean synaptic input into the proportion of neurons driven above threshold, while a separate factor accounts for the neurons that are not refractory. These constitute the key elements for the population equations to provide a coarse-grained description of neural populations, inspired by the single-neuron mechanisms introduced in this chapter.

Chapter 3

Localized Wilson-Cowan model

Modeling every single neuron and synapse in the brain is often not tractable because the number of cells involved in sensory processing and higher functions is too vast. Furthermore, many neural tasks are based on global processes. Therefore, studying only micro-properties (single cells) may fail to provide insight into how the brain performs. Meso-scale approaches are useful for computational studies on global processes in the brain. On the computational side, low-dimensional approximations of meso-scale neural dynamics are achieved via so called *mean-field* models; see Table 3.1. These are the models which are based on variables describing the population dynamics of neurons at a coarse-grained continuum limit.

Table 3.1: **Categories of mean-field models.**

Type	**Domain**	**Neural activity**	**Examples**
Localized Wilson-Cowan	Time	Oscillations Multi-stability Temporal dynamics	Wilson-Cowan 1972 [12]
Neural mass models	Time	Oscillations Multi-stability Temporal dynamics	Jansen-Rit 1995 [9]
Neural field models	Time-space	Travelling waves Pattern formation (bumps) Seizure propagation	Wilson-Cowan 1973 [13] Amari 1977 [1]
Biophysical (derived) mean-field	Time Time-space	Asynch. irregular states Up-down states Balanced exc-inh regimes	LIF mean-field: Brunel-Hakim 1999 [3] AdEx mean-field: di Volo et al. 2021 [5]

A key hypothesis regarding the mean-field models is that, while local interactions between individual neurons may be mostly random, this randomness induces precise long-range interactions on average. This hypothesis supports the idea of modeling neural populations in terms of coarse-grained variables defined at a continuum limit. The model established by Hugh R.

Wilson and Jack D. Cowan in 1972, is precisely based on such an idea. This model represents a foundational shift in computational neuroscience from single-cell modeling to the analysis of neural populations. By employing coarse-grained approximations and nonlinear differential equations, the model describes the averaged dynamics of interacting excitatory and inhibitory neurons. This shift from microscopic to macroscopic dynamics makes it possible to analyze the behavior of neural populations using nonlinear dynamical systems.

The model provides a computationally light, low-dimensional alternative to biophysically detailed network models, which are often too high-dimensional to manipulate effectively. This makes the neural dynamics generated by the model more tractable.

The Wilson-Cowan model is a localized model. That is, it neglects any extension of the populations in space. It focuses on temporal phenomena, assuming dense, random connections within a population in close spatial proximity.

The model successfully predicts complex physiological behaviors, including multiple stable states, hysteresis (a potential basis for short-term memory), and limit cycle oscillations (neural rhythms). Extensions of the model accurately predict the propagation speed of migraine-related seizure dynamics and describe the transition from baseline neural activity to hypersynchronization.

In the following sections, we will look into the details of the Wilson-Cowan localized neural field model.

3.1 Modeling assumptions

The Wilson-Cowan model specifically treats localized populations, concentrated in one region. It is based on the following assumptions:

Close proximity: Neurons are spatially close and densely connected. Therefore, we can neglect spatial interactions, and focus only on how activity changes over time (t), ignoring the space variable (x).

Proportions of activity: Activity is represented by the proportion of firing cells in the population rather than individual action potentials.

Excitatory-inhibitory interactions: The model assumes that all complex neural processes depend on the interactions between an excitatory population and an inhibitory population. Excitatory neurons increase the probability that other neurons fire. On the contrary, inhibitory neurons decrease the probability that other neurons fire. The balance between these two populations governs the cortical dynamics.

3.2 Mathematical building blocks

The model is composed of two coarse-grained state variables denoted by $\bar{E}(t)$ and $\bar{I}(t)$. They represent instantaneous proportions of firing neurons in the excitatory population ($\bar{E}(t)$) and inhibitory population ($\bar{I}(t)$), respectively. Since these variables represent proportions, we have $0 \leq \bar{E}(t), \bar{I}(t) \leq 1$ for all $t \geq 0$. The model equations are given as follows:

$$\begin{aligned}\mu\frac{\mathrm{d}\bar{E}}{\mathrm{d}t} &= -\bar{E}(t) + \big(1-\bar{E}(t)\big)\,S_{\mathrm{e}}\big(w_{\mathrm{ee}}\,\bar{E}(t) - w_{\mathrm{ei}}\bar{I}(t) + \bar{P}(t)\big),\\ \mu\frac{\mathrm{d}\bar{I}}{\mathrm{d}t} &= -\bar{I}(t) + \big(1-\bar{I}(t)\big)\,S_{\mathrm{i}}\big(w_{\mathrm{ie}}\bar{E}(t) - w_{\mathrm{ii}}\bar{I}(t) + \bar{Q}(t)\big),\end{aligned} \tag{3.1}$$

Here $\mu > 0$ is the time constant, S_{e}, S_{i} are the population transfer functions of the excitatory and inhibitory populations, respectively, and w_{ee}, w_{ei}, w_{ie}, $w_{\mathrm{ii}} \geq 0$ denote the connection weights[1]. Here $\bar{P}$ and $\bar{Q}$ denote the (possibly) time-dependent external input to the excitatory and inhibitory populations, respectively. The model equations are initialized from properly chosen initial conditions $0 \leq \bar{E}(0),\, \bar{I}(0) \leq 1$.

In the model equations, we observe two components: the negative linear feedback terms $-\bar{E}$, $-\bar{I}$ and the synaptic interactions $(1-\bar{E})\,S_{\mathrm{e}}$ and $(1-\bar{I})\,S_{\mathrm{i}}$. The former guarantees that the populations converge to the resting state value (typically 0) if there are no synaptic or external inputs to the populations. The latter provide the proportions of activated (firing) neurons due to the input that the populations receive.

A population transfer function provides the proportion of the activated neurons in its corresponding population, with respect to the input (both synaptic and external input) that the population receives. It assumes that all the neurons in the population are excitable, ready to produce action potentials. In other words, it neglects the refractoriness of the neurons in the population (see Section 2.3 for the refractoriness at the level of a neuron). Here the multiplication with $\big(1-\bar{E}(t)\big)$ and $\big(1-\bar{I}(t)\big)$ in the excitatory and inhibitory population equations, respectively, fixes this. These multiplications are introduced in the model to take into account the refractoriness of the neurons in excitatory and inhibitory populations at time t.

3.2.1 Refractoriness

After firing, a neuron enters a refractory period where it cannot fire again immediately, as explained in Section 2.3. This property is not taken into account by the transfer function, which assumes that all neurons are ready to fire, ignoring the refractory period of the neurons. To rectify this, one should introduce the multiplication factors $\big(1-\bar{E}(t)\big)$ and $\big(1-\bar{I}(t)\big)$ in (3.1).

Let us recall that the terms $\bar{E}(t)$ and $\bar{I}(t)$ represent the proportions of firing neurons in excitatory and inhibitory populations at instant t, respectively. Let us assume that the refractory period has a duration of r for all neurons. Then, the proportions of the excitatory and inhibitory neurons which are refractory at instant t are found respectively by

$$\int_{t-r}^{t} \bar{E}(t')\,\mathrm{d}t', \qquad \int_{t-r}^{t} \bar{I}(t')\,\mathrm{d}t'.$$

Therefore, the proportions of the nonrefractory neurons at instant t are

$$1 - \int_{t-r}^{t} \bar{E}(t')\,\mathrm{d}t', \qquad 1 - \int_{t-r}^{t} \bar{I}(t')\,\mathrm{d}t'.$$

[1]The subscripts denote which population provides the synaptic input to which population. It should be read from right to left, e.g. w_{ei} is the connection weight corresponding to the interactions where the inhibitory population provides the synaptic input to the excitatory population.

The integral shows in each population, the proportion of the neurons which fired in the $r-$length time interval which preceded the time instant t. It is the proportion of the neurons which are in the refractory period, thus which cannot fire at the instant t. Its subtraction from the total proportion 1 provides the proportion of the neurons which are ready to fire once there is sufficient stimulation.

3.2.2 Population transfer functions

A neuron produces spikes, i.e. it fires, if it receives a sufficiently strong excitatory synaptic input, and if it is not in its refractory period. According to the assumption of close proximity (see Sec. 3.1), the neurons in a population are densely distributed in a small region. This allows us to suppose that every neuron receives the same synaptic input on average in the population.

The transfer function of a population gives the proportion of firing neurons as output, with respect to the synaptic input that the population receives. It neglects the refractory period of the neurons. In other words, it considers the neurons in the population as if they were ready to fire no matter when they fired the last time. This is rectified by the multiplication factors explained in Sec. 3.2.1.

The general form of transfer functions can be derived by assuming that the individual firing threshold θ of the neurons within a population are characterized by a distribution function $D(\theta)$. One typical choice for $D(\theta)$ is a unimodal distribution, e.g. a standard normal distribution. Assuming further that each neuron receives the same number of excitatory and inhibitory afferent inputs, all neurons will, on average, experience the same mean excitation ξ. Consequently, the population transfer function $S(\xi)$ can be expressed as a function

- that is monotonically increasing on $(-\infty, \infty)$;
- that satisfies $\lim_{\xi\to-\infty} = 0$ and $\lim_{\xi\to\infty} = 1$,
- that has only one inflection point.

A common choice for a transfer function is the sigmoid function, see Figure 3.1. It is given by

$$S(\xi) = \frac{1}{1 + \mathrm{e}^{-\beta\,(\xi-\xi_0)}},$$

where $\beta > 0$ determines the sharpness of the nonlinearity, which represents the activity threshold heterogeneity of the population, and ξ_0 determines the population activity threshold. We will denote the transfer functions of the excitatory and inhibitory populations by S_e and S_i, respectively.

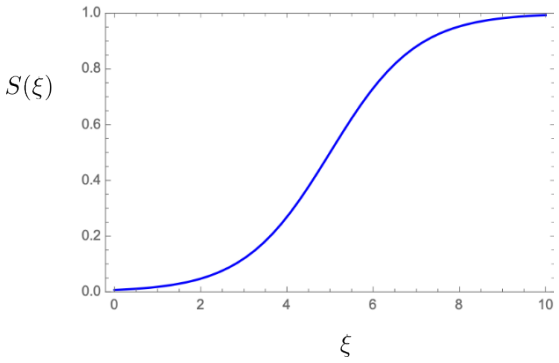


Figure 3.1: **Sigmoid function.**

Physiological interpretation of the sigmoid function

The sigmoidal shape of the population transfer function has a natural physiological interpretation. Recall that $S(\xi)$ represents the proportion of neurons in the population whose firing threshold is exceeded when the population receives a mean input ξ. The nonlinear shape of S therefore reflects the fact that neurons within the population do not necessarily have identical firing thresholds. For sufficiently small values of the input ξ, the synaptic excitation received by the population is below the firing threshold of almost all neurons. Consequently, only a very small proportion of neurons is activated and

$$S(\xi) \approx 0.$$

As the input increases, the firing thresholds of an increasing number of neurons are exceeded. The proportion of activated neurons therefore increases rapidly in the intermediate region of the sigmoid. Finally, for sufficiently large input, the firing threshold of almost every neuron in the population is exceeded. Since $S(\xi)$ represents a proportion, it cannot increase indefinitely, and hence it saturates:

$$S(\xi) \longrightarrow 1 \qquad \text{as} \qquad \xi \longrightarrow +\infty.$$

Thus, the lower and upper plateaus of the sigmoid correspond, respectively, to the regimes in which almost none and almost all of the excitable neurons respond to the input. The parameter ξ_0 determines the location of the transition between these two regimes. Indeed,

$$S(\xi_0) = \frac{1}{2},$$

so that ξ_0 may be interpreted as the input level for which half of the population is activated. The parameter β controls the steepness of this transition. Differentiating S gives

$$\frac{\mathrm{d}S}{\mathrm{d}\xi} = \beta S(\xi)\big(1 - S(\xi)\big),$$

and therefore

$$\frac{\mathrm{d}S}{\mathrm{d}\xi}\Big|_{\xi_0} = \frac{\beta}{4}.$$

Hence, large values of β correspond to a sharp transition between low and high population activity, whereas small values of β correspond to a gradual transition. From the population

point of view, a gradual transition is consistent with a larger heterogeneity of neural firing thresholds, while a sharp transition corresponds to a population whose neurons have more similar thresholds. The use of a sigmoid is therefore not merely a mathematical convenience. Sigmoid-like input-output relations are also consistent with experimentally observed population responses: weak inputs produce little activity, intermediate inputs generate a rapidly increasing response, and sufficiently strong inputs eventually lead to saturation; see Figure 3.2. The sigmoid provides a simple phenomenological representation of these three regimes.

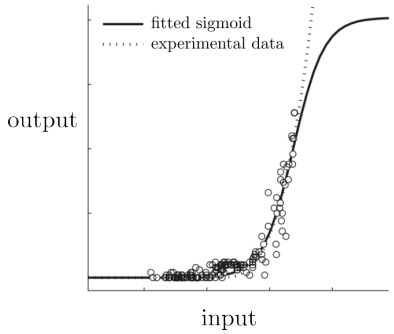


Figure 3.2: **Experimentally observed input-output curve of a neuron (adapted from [7]).**

3.2.3 From synaptic input to population activity

We now use the transfer function to construct an equation describing the temporal evolution of the activity of a localized neural population. Individual neurons receive many excitatory and inhibitory synaptic inputs. These inputs generate postsynaptic potentials which are integrated by the neuron over time, as explained in Chapter 2. At the population level, we replace these microscopic contributions by their average effect. Let $E(t)$ and $I(t)$ denote the instantaneous activities of the excitatory and inhibitory populations, respectively. The mean input arriving at an excitatory neuron can then be decomposed into three contributions:

$$w_1 E(t) - w_2 I(t) + P(t),$$

where $w_1, w_2 \geq 0$ denote the average strengths of excitatory and inhibitory interactions, respectively, and $P(t)$ denotes the external input to the excitatory population. The minus sign in front of the inhibitory contribution expresses the opposite effects of excitatory and inhibitory synaptic inputs. The same holds for the mean input arriving at an inhibitory neuron:

$$w_3 E(t) - w_4 I(t) + Q(t).$$

Synaptic inputs do not act instantaneously and disappear immediately. A postsynaptic potential produced at an earlier time t' may still contribute to the membrane response at the present time t. We describe this temporal effect by a synaptic kernel $\alpha(t - t')$. The effective mean inputs received by the excitatory and inhibitory populations at time t are therefore written

respectively as

$$\begin{aligned}\xi(t) &= \int_{-\infty}^{t} \alpha(t-t')\Big(w_1 E(t') - w_2 I(t') + P(t')\Big)\,\mathrm{d}t',\\ \eta(t) &= \int_{-\infty}^{t} \alpha(t-t')\Big(w_3 E(t') - w_4 I(t') + Q(t')\Big)\,\mathrm{d}t'.\end{aligned} \tag{3.2}$$

The kernel α specifies how the influence of an input changes with the time elapsed since its arrival. Consequently, (3.2) expresses the fact that the current state of a neuron depends not only on the input arriving exactly at time t, but also on its recent synaptic history. The transfer functions convert these effective inputs into the fractions of neurons whose firing thresholds are exceeded:

$$S_{\mathrm{e}}\big(\xi(t)\big), \qquad S_{\mathrm{i}}\big(\eta(t)\big).$$

These quantities alone, however, are not yet the actual population activity induced by the arriving input to the population. This is because some of the neurons whose thresholds are exceeded may be refractory. Suppose again that the refractory period has duration r. The proportions of excitatory and inhibitory neurons which have fired during the interval $[t-r,\, t]$ are

$$\int_{t-r}^{t} E(t')\,\mathrm{d}t', \qquad \int_{t-r}^{t} I(t')\,\mathrm{d}t'.$$

Accordingly, the proportions of excitatory and inhibitory neurons available to fire at time t are

$$1 - \int_{t-r}^{t} E(t')\,\mathrm{d}t', \qquad 1 - \int_{t-r}^{t} I(t')\,\mathrm{d}t'.$$

A neuron contributes to the new population activity only if two conditions are simultaneously satisfied: it must be nonrefractory and its firing threshold must be exceeded by the effective synaptic input. Combining these two requirements gives

$$\begin{aligned}E(t+\mu) &= \left(1 - \int_{t-r}^{t} E(t')\,\mathrm{d}t'\right) S_{\mathrm{e}}\left(\int_{-\infty}^{t} \alpha(t-t')\Big(w_1 E(t') - w_2 I(t') + P(t')\Big)\,\mathrm{d}t'\right),\\ I(t+\mu) &= \left(1 - \int_{t-r}^{t} I(t')\,\mathrm{d}t'\right) S_{\mathrm{i}}\left(\int_{-\infty}^{t} \alpha(t-t')\Big(w_3 E(t') - w_4 I(t') + Q(t')\Big)\,\mathrm{d}t'\right).\end{aligned} \tag{3.3}$$

Here $\mu > 0$ represents a characteristic time required for the population to respond to its input. We find a particularly useful physiological interpretation in (3.3): the activity observed at the subsequent time $t+\mu$ is determined by the proportion of neurons that are both available to fire and sufficiently stimulated to fire. Schematically,

$$\boxed{\text{population activity} = \text{available fraction} \times \text{fraction above threshold.}}$$

In (3.3), the first factor accounts for refractoriness, whereas the second factor accounts for the nonlinear response of the population to its synaptic input. Thus, at this stage, the localized Wilson-Cowan description contains two physiologically important temporal forms. The first is the integral over the preceding refractory interval, which accounts for the recent firing history of the population. The second is the convolution with α, which accounts for the temporal integration of postsynaptic potentials.

Although (3.3) has a direct physiological interpretation, it is an equation with integrals and delays. Therefore, it is considerably more complicated to analyze than the ordinary differential

equations introduced in (3.1). The next step is to introduce an appropriate coarse-graining procedure. This will replace the rapidly varying instantaneous population activity by a temporally averaged activity and simplify the equations with integrals and delays. This simplification will lead to the Wilson-Cowan equations used throughout the remainder of this chapter.

3.3 Time coarse-graining

The equations in (3.3) keep track of two kinds of recent history. The refractory integral counts the neurons that have fired recently, while the convolution with α describes how earlier synaptic input still affects a neuron. These equations are close to the physiological description, but the integrals and time delays make them difficult to study. Wilson and Cowan therefore replaced rapidly changing activities by short-time averages to simplify these equations [12].

For any time-dependent quantity f, we can define its moving average over a window of duration $s > 0$ by

$$\bar{f}_s(t) = \frac{1}{s}\int_{t-s}^{t} f(t')\,\mathrm{d}t'.$$

The average retains the changes that are slow and that take longer time compared with s, while reducing the variations that are much faster and that happen quickly within the time window s. The choice of s is therefore part of the approximation: a very long window would also remove changes that we want the model to describe.

3.3.1 The refractory and synaptic time windows

Choosing $s = r$, where r denotes the refractory period, gives an exact rewriting of the refractory terms:

$$\int_{t-r}^{t} E(t')\,\mathrm{d}t' = r\bar{E}_r(t), \qquad \int_{t-r}^{t} I(t')\,\mathrm{d}t' = r\bar{I}_r(t). \tag{3.4}$$

Here E and I are instantaneous firing activity per unit time, whereas $r\bar{E}_r$ and $r\bar{I}_r$ are dimensionless fractions of cells that fired during the preceding refractory interval. The fractions available to fire are consequently $1 - r\bar{E}_r$ and $1 - r\bar{I}_r$. This is why refractoriness multiplies the response functions.

The synaptic kernel $\alpha(u)$ describes the effect of an input u, some units of time after its arrival. This is because the effect of an input does not occur instantly when the input arrives at the neuron. The synaptic kernel $\alpha(u)$ describes how the effective input decays after the instant that the input arrives at the neuron. For example, a decaying kernel can be chosen as $\alpha(u) = \exp(-u/\mu)$ for $u \geq 0$, where τ_s is a synaptic decay time. If the mean input changes slowly over this decay time, its convolution can be approximated by a short-time average multiplied by the area of the kernel:

$$\int_{-\infty}^{t} \alpha(t-t')X(t')\,\mathrm{d}t' \;\approx\; \kappa\,\bar{X}(t), \qquad \kappa = \int_{0}^{\infty} \alpha(u)\,\mathrm{d}u. \tag{3.5}$$

For this specific example, we choose $\kappa = \mu$. A rectangular kernel of height one and width κ has the same area and gives an intuitive picture of the approximation. In the following, we

use one short-time average for the activity and input variables. This will provide a further coarse-graining approximation.

Write $\bar{E}$, $\bar{I}$, $\bar{P}$, and $\bar{Q}$ for these averages. The integrals in (3.4) and (3.5) turn the physiological equations in (3.3) approximately into

$$\begin{aligned} E(t+\mu) &\approx (1-r\bar{E})\, S_{\mathrm{e}}\big(\kappa(w_1\bar{E} - w_2\bar{I} + \bar{P})\big)\,, \\ I(t+\mu) &\approx (1-r\bar{I})\, S_{\mathrm{i}}\big(\kappa(w_3\bar{E} - w_4\bar{I} + \bar{Q})\big)\,. \end{aligned} \tag{3.6}$$

All averages on the right-hand side are evaluated at time t. Each nonnegative coefficient w_j measures an average synaptic connection strength. The minus signs in front of $w_2\bar{I}$ and $w_4\bar{I}$ represent inhibitory input.

3.3.2 A differential equation for the averaged activity

After averaging, we describe the left-hand sides by the slowly varying quantities $\bar{E}$ and $\bar{I}$. If these quantities have a small change during a small time increase μ, a first-order Taylor approximation[2] gives

$$\bar{E}(t+\mu) \approx \bar{E}(t) + \mu\frac{\mathrm{d}\bar{E}}{\mathrm{d}t}, \qquad \bar{I}(t+\mu) \approx \bar{I}(t) + \mu\frac{\mathrm{d}\bar{I}}{\mathrm{d}t}.$$

We substitute it into (3.6) and rearrange the terms to obtain

$$\begin{aligned} \mu\frac{\mathrm{d}\bar{E}}{\mathrm{d}t} &\approx -\bar{E} + (1-r\bar{E})\, S_{\mathrm{e}}\big(\kappa(w_1\bar{E} - w_2\bar{I} + \bar{P})\big)\,, \\ \mu\frac{\mathrm{d}\bar{I}}{\mathrm{d}t} &\approx -\bar{I} + (1-r\bar{I})\, S_{\mathrm{i}}\big(\kappa(w_3\bar{E} - w_4\bar{I} + \bar{Q})\big)\,, \end{aligned}$$

The negative terms $-\bar{E}$ and $-\bar{I}$ comes from moving $\bar{E}(t)$ and $\bar{I}(t)$ to the right-hand side of the Taylor approximation.

In many cases, we can choose $r = 1$. This choice is also supported by experimental studies showing that the refractory period is around 1-2 ms [6]. Moreover, we can consider the factor κ as absorbed into the connection weights. For instance, the effective excitatory-to-excitatory weight is κw_1. We can write this effective weight as $w_{\mathrm{ee}} = \kappa w_1$. This gives exactly the form introduced in (3.1):

$$\begin{aligned} \mu\frac{\mathrm{d}\bar{E}}{\mathrm{d}t} &= -\bar{E} + (1-\bar{E})S_{\mathrm{e}}(w_{\mathrm{ee}}\bar{E} - w_{\mathrm{ei}}\bar{I} + \bar{P}), \\ \mu\frac{\mathrm{d}\bar{I}}{\mathrm{d}t} &= -\bar{I} + (1-\bar{I})S_{\mathrm{i}}(w_{\mathrm{ie}}\bar{E} - w_{\mathrm{ii}}\bar{I} + \bar{Q}). \end{aligned} \tag{3.7}$$

This approximation is useful for population changes slower than the refractory and synaptic time scales. It does not retain individual spikes or every rapid fluctuation.

[2] For a sufficiently smooth function f, the Taylor formula for a small increment $\mu \ll 1$ about t is $f(t+\mu) = f(t) + \mu f'(t) + O(\mu^2)$. A first-order approximation neglects the terms of order μ^2 and higher. Here f' denotes $\frac{\mathrm{d}f}{\mathrm{d}t}$ and f denotes either $\bar{E}$ or $\bar{I}$.

3.4 Reading the localized equations

The two equations in (3.7) have the same structure. Each contains a loss term ($-\bar{E}$ or $-\bar{I}$), an available fraction ($1-\bar{E}$ or $1-\bar{I}$), and a nonlinear response to the total input. For the excitatory population, the total input is $w_{\mathrm{ee}}\bar{E} - w_{\mathrm{ei}}\bar{I} + \bar{P}$: recurrent excitation $w_{\mathrm{ee}}\bar{E}$ raises it, inhibition $-w_{\mathrm{ei}}\bar{I}$ lowers it, and $\bar{P}$ is external stimulation. The inhibitory equation is read in the same way, using $w_{\mathrm{ie}}\bar{E} - w_{\mathrm{ii}}\bar{I} + \bar{Q}$. The model is localized, i.e. its variables depend on time but have no spatial coordinate since all neurons in the populations are assumed to be located in the same region.

A *steady state* is a pair $(\bar{E}_*, \bar{I}_*)$ that does not change with time, that is

$$\mu\frac{\mathrm{d}\bar{E}_*}{\mathrm{d}t} = \mu\frac{\mathrm{d}\bar{I}_*}{\mathrm{d}t} = 0.$$

It satisfies

$$\bar{E}_* = (1-\bar{E}_*)S_{\mathrm{e}}(w_{\mathrm{ee}}\bar{E}_* - w_{\mathrm{ei}}\bar{I}_* + \bar{P}),$$
$$\bar{I}_* = (1-\bar{I}_*)S_{\mathrm{i}}(w_{\mathrm{ie}}\bar{E}_* - w_{\mathrm{ii}}\bar{I}_* + \bar{Q}),$$

when the external inputs are constant. A resting state means that activity stays at such a low, stable level. It need not mean that every neuron is silent. Nevertheless, in general, the parameters of sigmoid response functions are chosen such that $S_{\mathrm{e}}(0) = S_{\mathrm{i}}(0) = 0$. Then, $(0,0)$ is a steady state without external input, as in the baseline neural activity convention discussed by Wilson and Cowan [12].

The equations also respect the meaning of activity as a fraction. If $0 \le S_{\mathrm{e}}, S_{\mathrm{i}} \le 1$ and the initial activities lie in $[0,1]$, the vector field points inward at both ends of this interval: at $\bar{E} = 0$ its derivative cannot be negative, while at $\bar{E} = 1$ it is negative. The same holds for $\bar{I}$. Hence solutions remain between 0 and 1.

3.5 Temporal activity patterns

Different connection strengths and inputs can produce different time courses. Figure 3.3 summarizes the three patterns shown in the lecture.

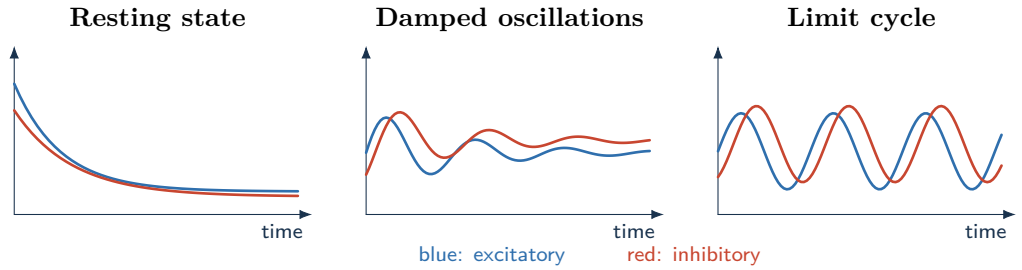


Figure 3.3: **Schematic temporal patterns of a localized excitatory-inhibitory model.** The curves illustrate qualitative behaviors of the model.

In a **resting state**, activity approaches a fixed level and remains there unless an input perturbs it, see the left hand-side plot in Figure 3.3. In **damped oscillations**, excitation and inhibition

temporarily rise and fall after a perturbation, but the fluctuations become smaller and the system eventually settles at a steady state, as shown in the plot in the middle in Figure 3.3. A **limit cycle** instead produces sustained, repeated changes of activity: after an initial transient, the oscillation continues rather than dying away, see the right hand-side plot in Figure 3.3.

3.6 Scope of the localized model

The localized Wilson-Cowan model uses two time-dependent variables to describe excitatory-inhibitory interactions. Depending on its parameters and inputs, activity can approach a steady level, respond transiently, or sustain a rhythm. The model has no spatial variable, so it cannot describe where activity occurs or how a wave travels across the cortex. Spatial neural-field models add position to the localized models and connections between locations. Therefore, the localized Wilson-Cowan model is useful for temporal dynamics and is a starting point for spatial extensions. In the next chapter, we will continue with a spatial extension of the localized Wilson-Cowan model.

Chapter 4

Spatially extended Wilson-Cowan model

4.1 Why add space?

The localized Wilson-Cowan model of Chapter 3 describes how the excitatory and inhibitory activities in one small region change with time. It can describe a resting state or a rhythm, but it cannot say *where* the activity occurs. A cortical response may instead be strong in one place, remain localized, or travel across tissue. To describe such phenomena, we allow the population activities and external inputs to depend on both position x and time t.

We use the notation $\bar{E}(x,t)$ and $\bar{I}(x,t)$ for the coarse-grained excitatory and inhibitory activities. For simplicity, x represents a position on a one-dimensional strip of cortex, which we will denote by $\Omega = [-x_0, x_0]$. It can be thought of as a cross-section of an approximately two-dimensional cortical sheet.

The main new ingredient in the spatially extended Wilson-Cowan model is lateral connectivity: neurons near x can receive input from neurons at other positions X. Consequently, the model couples activity at different points of the tissue.

4.2 Modeling assumptions and activity variables

As in Chapter 3, we distinguish excitatory and inhibitory neurons, and use population averages instead of tracking single action potentials. The two neuron types are assumed to be homogeneously distributed throughout the cortical tissue. Neural connections of all four types are allowed: from excitatory to excitatory (ee), from inhibitory to excitatory (ei), from excitatory to inhibitory (ie), and from inhibitory to inhibitory (ii). We assume that the tissue is homogeneous and isotropic: the same connection rule applies at each position and in all directions. One should note that these assumptions describe an idealized setting, not every detail of real cortex.

It is useful to distinguish instantaneous neural activity from its smoother, time-averaged representation. Let $E(x,t)$ and $I(x,t)$ describe the instantaneous proportion of firing neurons in the two populations at x. In the equations that we will see below, they represent the proportion of the firing neurons per unit time. Therefore, their integral over a short time interval is a fraction of neurons that fired. This fraction is considered with respect to the corresponding population. The final variables $\bar{E}(x,t)$ and $\bar{I}(x,t)$ are normalized coarse-grained activities. With this convention $0 \leq \bar{E}, \bar{I} \leq 1$ represent the active fractions of the excitatory and inhibitory populations. The bar therefore matters: $E(x,t)$ is not identical to $\bar{E}(x,t)$, and $I(x,t)$ is not identical to $\bar{I}(x,t)$

The time-averaged external inputs to the two populations are denoted by $\bar{P}(x,t)$ and $\bar{Q}(x,t)$. As before, S_e and S_i convert effective input into population response. We take them to be nondecreasing and bounded functions, often chosen to have a sigmoid shape.

4.3 Distance-dependent neural connectivity

We introduce a nonnegative kernel $\omega_{ab}(d)$ for each type of neural connection. The first subscript a names the receiving population and the second subscript b the sending population. Thus, $\omega_{\mathrm{ei}}(x-X)$ measures how strongly inhibitory activity at X contributes to the input of excitatory neurons at x. The argument $d = x - X$ provides the distance between the two locations. We assume that the connectivity is symmetric, i.e. $\omega_{ab}(d) = \omega_{ab}(-d)$.

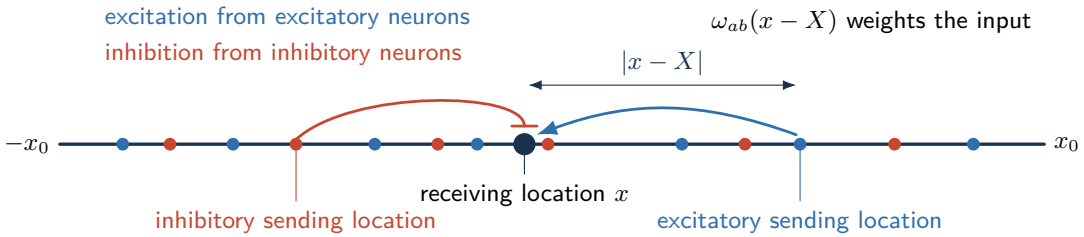


Figure 4.1: **Spatial coupling along a cortical strip.** Excitatory and inhibitory populations coexist along the tissue. Activity at a sending position X contributes to the input at a receiving position x according to the distance $|x-X|$ and the appropriate connectivity kernel $\omega_{ab}(x-X)$.

Consider an excitatory sender in a small interval of width $\mathrm{d}X$ around X. Its contribution to the excitatory population at the position x and at the instant t is proportional to

$$E(X,t)\,\omega_{\mathrm{ee}}(x-X)\,\mathrm{d}X.$$

Adding the contributions from all sending positions gives a spatial integral. For a function F and a kernel ω_{ab} we write

$$(F * \omega_{ab})(x,t) := \int_{-x_0}^{x_0} F(X,t)\,\omega_{ab}(x-X)\,\mathrm{d}X.$$

This is a *spatial convolution* on the modeled cortical strip. In particular, $(E*\omega_{\mathrm{ee}})(x,t)$ represents excitation received by the excitatory population, while $(I*\omega_{\mathrm{ei}})(x,t)$ represents inhibition. The inhibitory population similarly receives $(E*\omega_{\mathrm{ie}})(x,t)$ and $(I*\omega_{\mathrm{ii}})(x,t)$.

The scale σ_{ab} of a kernel sets the connectivity range. Its amplitude sets the connection strength. Two common profiles are

$$\omega_{ab}(d) = c_{ab}\exp(-|d|/\sigma_{ab}) \quad \text{(exponential)},$$
$$\omega_{ab}(d) = c_{ab}\exp[-(d/\sigma_{ab})^2] \quad \text{(Gaussian)}.$$

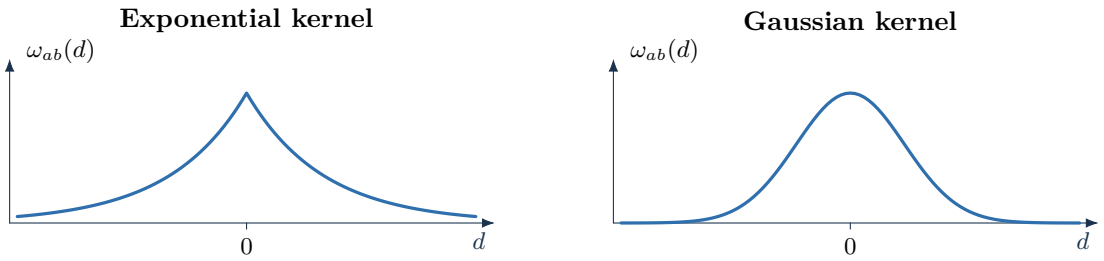


Figure 4.2: **Examples of distance-dependent connectivity kernels.** Both curves peak at zero separation and decrease with distance. Their relative widths and heights depend on the chosen values of σ_{ab} and c_{ab}.

Both give stronger connections between nearby locations and weaker connections between distant locations. Here $c_{ab} \geq 0$ and $\sigma_{ab} > 0$. Different connection types can have different strengths and ranges. We keep inhibitory kernels nonnegative and place the inhibitory minus sign explicitly in the equations, so that the signs are easy to read.

An action potential takes time to travel from X to x. If its propagation speed is ν, the input arriving at the population at time t depends on the presynaptic activity at the earlier time $t - \dfrac{|x-X|}{\nu}$. A delayed excitatory contribution is therefore

$$\int_{-x_0}^{x_0} E\left(X, t - \frac{|x-X|}{\nu}\right) \omega_{\mathrm{ee}}(x-X)\,\mathrm{d}X. \tag{4.1}$$

The same replacement applies to the other three connection types. Following the simplification made in the localized model explained in Chapter 3, we neglect this conduction delay by assuming that the propagation speed ν is very large, i.e. $\dfrac{|x-X|}{\nu} \approx 0$. This allows us to simplify (4.1) as

$$\int_{-x_0}^{x_0} E(X,t)\,\omega_{\mathrm{ee}}(x-X)\,\mathrm{d}X. \tag{4.2}$$

4.4 From temporal history to field equations

The derivation follows the same two biological ideas as the localized model. First, postsynaptic effects persist for a short time, so inputs are integrated over recent history. Second, a neuron that has just fired is temporarily refractory.

Let the mean input to the excitatory population **before** temporal effects be

$$U_{\mathrm{e}}(x,t) = (E * \omega_{\mathrm{ee}})(x,t) - (I * \omega_{\mathrm{ei}})(x,t) + P(x,t).$$

We define the mean input U_{i} to the inhibitory population in a similar way, by replacing the pair of kernels by ω_{ie} and ω_{ii} and P by Q:

$$U_{\mathrm{i}}(x,t) = (E * \omega_{\mathrm{ie}})(x,t) - (I * \omega_{\mathrm{ii}})(x,t) + Q(x,t).$$

Let $\alpha(s)$ describe how much of a synaptic input remains after elapsed time $s \geq 0$. It is a decaying kernel and it models that the input becomes weaker as time passes. A typical choice is exponential decay: $\alpha(s) = \exp(-s/\mu)$ where $\mu > 0$ denotes the decay time scale. The effective excitatory input atb x is modeled as

$$\xi_{\mathrm{e}}(x,t) = \int_{-\infty}^{t} \alpha(t-t')\, U_{\mathrm{e}}(x,t')\, \mathrm{d}t'.$$

The effective inhibitory input is written in the same way once we replace U_{e} with U_{i}:

$$\xi_{\mathrm{i}}(x,t) = \int_{-\infty}^{t} \alpha(t-t')\, U_{\mathrm{i}}(x,t')\, \mathrm{d}t'.$$

If the refractory period lasts $r > 0$, at the location x, the fraction of excitatory and inhibitory neurons that have fired within the interval of length r preceding the instant t is found respectively via the following integrals:

$$\int_{t-r}^{t} E(x,t')\, \mathrm{d}t', \qquad \int_{t-r}^{t} I(x,t')\, \mathrm{d}t'$$

The same holds for inhibitory neurons. This implies that the fractions of excitatory and inhibitory neurons available to fire at (x,t) are

$$1 - \int_{t-r}^{t} E(x,t')\, \mathrm{d}t', \qquad 1 - \int_{t-r}^{t} I(x,t')\, \mathrm{d}t'.$$

Combining availability with the transfer functions gives

$$\begin{aligned} E(x,t+\tau) &= \left(1 - \int_{t-r}^{t} E(x,t')\, \mathrm{d}t'\right) S_{\mathrm{e}}\big(\xi_{\mathrm{e}}(x,t)\big), \\ I(x,t+\tau) &= \left(1 - \int_{t-r}^{t} I(x,t')\, \mathrm{d}t'\right) S_{\mathrm{i}}\big(\xi_{\mathrm{i}}(x,t)\big). \end{aligned} \tag{4.3}$$

where τ represents a short activation delay. We choose $\tau = \mu$. Here (4.3) has the same interpretation as its localized counterpart found in (3.6): updated activity is equal to the available fraction of neurons times the response to effective input. Their extra spatial integrals account for neurons at other locations.

To obtain simpler equations, we average rapid firing fluctuations in time. For the exponential synaptic kernel $\alpha(t-t') = \exp(-(t-t')/\mu)$, we define

$$\widehat{E}(x,t) \coloneqq \frac{1}{\mu} \int_{-\infty}^{t} E(x,t') \exp\left(-\frac{t-t'}{\mu}\right) \mathrm{d}t', \qquad \widehat{I}(x,t) \coloneqq \frac{1}{\mu} \int_{-\infty}^{t} I(x,t') \exp\left(-\frac{t-t'}{\mu}\right) \mathrm{d}t'. \tag{4.4}$$

We average P and Q similarly. Since the temporal averages in (4.4) act on t and the connectivity kernels in the spatial convolutions given in (4.2) act on x, their integrals can be interchanged.

Two useful simplifications now follow. Differentiating $\widehat{E}(x,t)$ and $\widehat{I}(x,t)$ with respect to t gives

$$E(x,t) = \mu \frac{\partial \widehat{E}}{\partial t}(x,t) + \widehat{E}(x,t), \qquad I(x,t) = \mu \frac{\partial \widehat{I}}{\partial t}(x,t) + \widehat{I}(x,t).$$

Moreover, if r is short relative to the changes retained by the average, then

$$\int_{t-r}^{t} E(x,t')\,\mathrm{d}t' \;\approx\; r\widehat{E}(x,t), \qquad \int_{t-r}^{t} I(x,t')\,\mathrm{d}t' \;\approx\; r\widehat{I}(x,t).$$

Finally, a first-order Taylor approximation in time replaces the activity a short time τ later by its present averaged value plus its time derivative as in the localized Wilson-Cowan model. These approximations remove the explicit temporal-history integrals from the update equations. The time averages still describe the approximate activity of interest, rather than individual spikes.

As in Chapter 3, we choose units and effective parameters that absorb the refractory factor r and the area μ of the synaptic kernel α. In particular, the normalized activities can be written as $\bar{E} = r\widehat{E}$ and $\bar{I} = r\widehat{I}$. The connection strengths and response functions are then considered as including the corresponding conversion factors. This leads to the standard, compact form of the spatially extended Wilson-Cowan equations:

$$\begin{aligned} \mu \frac{\partial \bar{E}(x,t)}{\partial t} &= -\bar{E}(x,t) + (1-\bar{E}(x,t))\, S_{\mathrm{e}}\big(J_{\mathrm{e}}(x,t)\big), \\ \mu \frac{\partial \bar{I}(x,t)}{\partial t} &= -\bar{I}(x,t) + (1-\bar{I}(x,t))\, S_{\mathrm{i}}\big(J_{\mathrm{i}}(x,t)\big), \end{aligned} \tag{4.5}$$

where the total inputs at position x are

$$\begin{aligned} J_{\mathrm{e}}(x,t) &= (\bar{E} * \omega_{\mathrm{ee}})(x,t) - (\bar{I} * \omega_{\mathrm{ei}})(x,t) + \bar{P}(x,t), \\ J_{\mathrm{i}}(x,t) &= (\bar{E} * \omega_{\mathrm{ie}})(x,t) - (\bar{I} * \omega_{\mathrm{ii}})(x,t) + \bar{Q}(x,t). \end{aligned}$$

4.5 Reading the spatial equations

At every location x, the first equation in (4.5) says that excitatory activity tends to decrease through $-\bar{E}$, while input excitation, input inhibition, and external input determine new activation. The factor $1-\bar{E}$ prevents more than the available fraction from being newly activated. The inhibitory equation has the same structure. If the response functions take values in $[0,1]$ and the initial normalized activities are in $[0,1]$, the dynamics point inward at the bounds 0 and 1 for each population, just as in the localized equations.

The equations are integro-differential: they contain time derivatives and spatial integrals. The spatial integral is the most important particularity compared to the localized model.

To solve the field equations one must specify the initial conditions $\bar{E}(x,0)$ and $\bar{I}(x,0)$, the external inputs, and the boundary conditions in space. Example boundary conditions are periodic boundaries (a mathematical strip with joined ends). The choice of boundary conditions can affect the observed spatial pattern.

4.6 Spatiotemporal activity patterns

The spatial model can distinguish several behaviors. A brief, localized stimulus may produce an *active transient*: activity rises and then returns to its background level. A *spatially inhomogeneous steady state* remains stronger at some positions than at others, even after the initial stimulus has ended. Activity may also oscillate in a restricted region, giving a spatially localized rhythm. Finally, a *travelling wave* is a profile that moves through the tissue; snapshots at successive times show its peak at different positions. These kinds of behavior arise for different parameter regimes and inputs in spatially extended Wilson-Cowan systems. They are possibilities of the model, not inevitable outcomes for every kernel. Figure 4.3 illustrates three of these cases.

To tell a travelling wave from a local oscillation, it is not enough to plot activity against time at one chosen x. We must compare the spatial profile at several times, or plot activity over both x and t. This is why the spatial extension is useful: it describes both *when* a population responds and *where* the response is located.

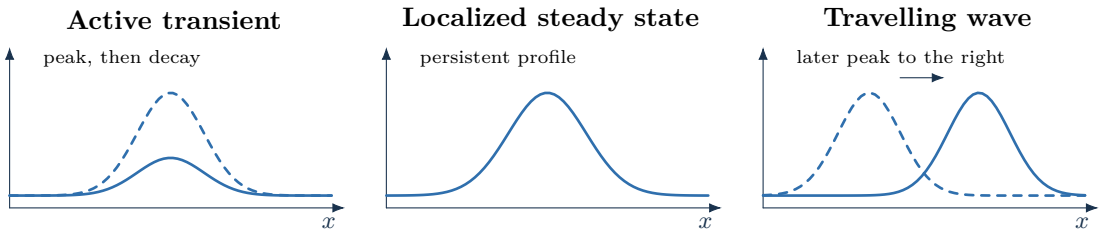


Figure 4.3: **Examples of spatial patterns of excitatory population activity.** Dashed and solid curves represent earlier and later snapshots of two different instants. From left to right: a transient fades, a steady profile stays in place, and a travelling profile moves.

4.7 Scope of the spatial model

The spatially-extende Wilson-Cowan equations provide a simple population-level description of interacting excitatory and inhibitory tissue. They preserve the response functions and refractoriness of the localized model, but add connections between positions. This makes it possible to study localized responses, persistent spatial patterns, and propagating activity. The model averages over individual neurons and neglects explicit axonal conduction delays. Its predictions depend on the chosen kernels, inputs, and spatial domain.

Chapter 5

Amari model

5.1 From two populations to a single population

The spatially extended Wilson-Cowan model in Chapter 4 uses one equation for excitatory population activity and another for inhibitory population activity. Amari's model uses a different, simpler description: one equation gives the average membrane potential at each position, while a single signed connectivity kernel describes both excitation and inhibition [1]. In this way, it encodes the excitation and inhibition in the connectivity kernel. This still allows us to ask whether an input disappears, produces activity everywhere, or leaves a localized region of activity after the input is removed.

We again consider a one-dimensional cross-section of neural tissue as in Chapter 4. Let $u(x,t)$ be the average membrane potential relative to a firing threshold at position x and time t. Thus $u(x,t) > 0$ means that the population at x is above threshold. Here zero is a convenient reference level, not a claim that the biological membrane potential is 0 mV. The one-dimensional cross-section can be represented as a finite interval $[-x_0, x_0]$, in particular in model simulations. For the model analysis below in Sections 5.4 and 5.5, we instead represent it as $[-\infty, \infty]$ to simplify the analysis. This avoids boundary effects and makes translation of a localized pattern particularly simple.

5.2 Modeling assumptions and field equation

We assume that the tissue is homogeneous: the connection from a sending position y to a receiving position x depends on their distance $|x - y|$, not on their absolute positions or on time. We also assume symmetric connections, $\omega(z) = \omega(-z)$, where ω denotes the connectivity kernel. In the Amari model, ω is chosen such that the model is restricted to populations of lateral inhibition type: i) Excitatory connections dominate the connectivity between proximate neurons, ii) Inhibitory connections dominate the connectivity between distant neurons. In this one-equation description, excitatory and inhibitory effects are not represented by separate state variables. They appear as positive and negative parts of ω, see the left panel of Figure 5.1.

The Amari equation is

$$\mu \frac{\partial u(x,t)}{\partial t} = -u(x,t) + \int_{-\infty}^{\infty} \omega(x-y) s\big(u(y,t)\big)\,\mathrm{d}y + h + P(x,t), \qquad \mu > 0. \tag{5.1}$$

The constant $\mu > 0$ is the time scale. The transfer function s converts membrane potential into population output response. The kernel ω weights this output according to distance between neurons. The constant h is the baseline input or threshold stimulus, usually chosen negative. External input $P(x,t)$ is an additional spatially and temporally varying stimulus. The integral represents the synaptic input arriving at x from all sending positions y. The term $-u$ makes the average potential relax toward h when synaptic input and external input P are absent, i.e. when $\mu \dfrac{\mathrm{d}u}{\mathrm{d}t} = -u + h$.

In the synaptic input, the order of operations matters. In Amari's equation, the sending population first produces $s(u(y,t))$, and then the kernel weights and sums these outputs. In the spatial Wilson-Cowan equations of Chapter 4, each population instead sums synaptic inputs before its population transfer function acts. Excitatory and inhibitory activities also have separate equations. Amari's u is a membrane-potential variable, whereas $\bar{E}$ and $\bar{I}$ in the Wilson-Cowan models are the fractions of neurons in each population that are available to fire. The two formulations therefore propose different coarse-grained descriptions.

5.3 Population transfer function and neural connectivity

For the analysis, we use the Heaviside function as the population transfer function:

$$s(u) = \begin{cases} 0, & u \le 0, \\ 1, & u > 0. \end{cases} \tag{5.2}$$

It says that a population produces an average output only when its average potential is above threshold, see the right panel of Figure 5.1. A good alternative could be a smoothly increasing and saturating function, such as a sigmoid function. However, the step function makes the boundaries of an activated region easy to track and therefore simplifies the model analysis as we will see in Section 5.4.

A standard example of a short-range-excitatory, long-range-inhibitory kernel is the Mexican-hat function:

$$\omega_\sigma(x) = \frac{2}{\sqrt{3}\,\sigma\pi^{1/4}} \left(1 - \frac{x^2}{\sigma^2}\right) \exp\!\left(-\frac{x^2}{2\sigma^2}\right), \qquad \sigma > 0. \tag{5.3}$$

Here $\sigma > 0$ is the scale. It determines the range of excitatory and inhibitory connections. The kernel ω_σ is positive (excitatory) for $|x| < \sigma$ and negative (inhibitory) for $|x| > \sigma$. Increasing σ spreads both parts over a wider spatial range. Another common choice as connectivity kernel is a narrow positive Gaussian minus a broader negative Gaussian. We will derive the equilibrium conditions for a generic symmetric short-range-excitatory, long-range-inhibitory kernel. These conditions are not tied to the particular formula in (5.3).

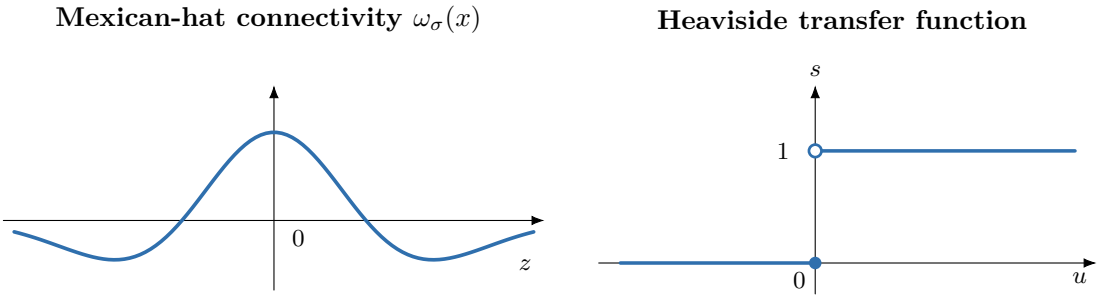


Figure 5.1: **Mexican-hat kernel and Heaviside function.** The signed Mexican-hat kernel gives positive influence from nearby positions and negative influence from more distant ones. The Heaviside step response selects active locations contributing to the synaptic input.

5.4 Equilibrium solutions

We now examine the equilibrium solutions of the Amari model equation. We work on the real line $x \in (-\infty, \infty)$. We set the external stimulus to zero, $P(x,t) = 0$ for all $x \in (-\infty, \infty)$ and $t \in [0, \infty)$. We write a time-independent solution as $u(x)$. At equilibrium, $\dfrac{\mathrm{d}u}{\mathrm{d}t} = 0$ must hold. Therefore, at equilibrium (5.1) becomes

$$u(x) = \int_{-\infty}^{\infty} \omega(x-y) s\big(u(y)\big)\,\mathrm{d}y + h.$$

The neurons at a location x are active when $u(x) > 0$ and inactive when $u(x) \leq 0$. We therefore define an active region by $(u) \coloneqq \{x \in \mathbb{R} : u(x,t) > 0\}$. The Heaviside function in (5.2) equals 1 on this region and 0 elsewhere. Therefore, we can write,

$$u(x) = \int_{R(u)} \omega(x-y) s\big(u(y)\big)\,\mathrm{d}y + h, \tag{5.4}$$

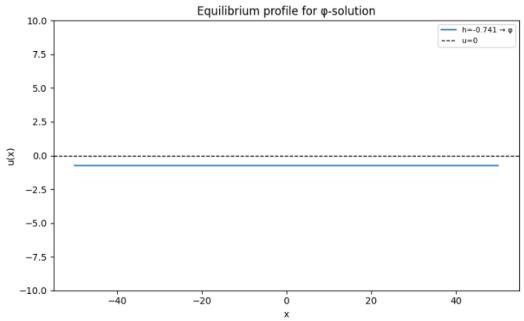
and since $s(u) = 1$ in $R(u)$,

$$u(x) = \int_{R(u)} \omega(x-y)\,\mathrm{d}y + h.$$

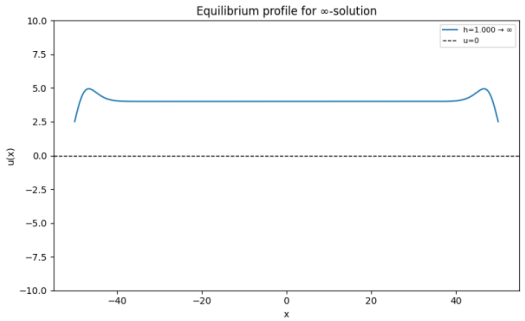
This restricted integral is the starting point of the equilibrium solution classification.

There are three basic kinds of equilibrium solutions. An *empty solution* has $R(u) = \varnothing$, so $u(x) \leq 0$ everywhere. An *all-active solution* has $R(u) = (-\infty, \infty)$, so $u(x) > 0$ everywhere. A *localized bump* has $R(u) = (x_1, x_2)$ for some $x_1 < x_2$, with width $a = x_2 - x_1 > 0$. The localized bump is nonpositive (≤ 0) outside that interval. Figure 5.2 gives the examples plots of these three equilibrium solution types. Empty, all-active and localized bump solutions are referred to as $\varnothing$-, ∞-, and a-solution, respectively.

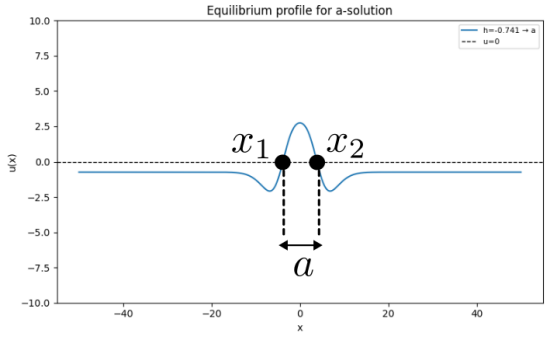
Since the connectivity kernel depends only on the difference $x - y$, translating a bump does not change the kernel. For a single bump, we may therefore shift the origin and take $R(u) = (0, a)$ without loss of generality.



(a) Empty solution



(b) All-active solution



(c) Localized bump of width a

Figure 5.2: **The three equilibrium-solution types.** The dashed horizontal line marks the threshold. The stationary profile is denoted by $u(x)$.

5.4.1 The kernel integral

We define the integral of the kernel of the Mexican-hat type (see Figure 5.1) and two measures that characterize its graph:

$$W(z) = \int_0^z \omega(x)\,\mathrm{d}x, \qquad W_m = \max_{z>0} W(z), \qquad W_\infty = \lim_{z\to\infty} W(z).$$

We have $W(0) = 0$. Since ω is symmetric (even), W is antisymmetric (odd): $W(-z) = -W(z)$. For a short-range-excitatory and long-range-inhibitory kernel of the type considered here, W rises as nearby excitation accumulates. It has one maximum W_m on the positive half-line, and then falls as inhibition is included. It converges to W_∞ as $z \to \infty$. Figure 5.3 provides an illustrative graph of $W(z)$.

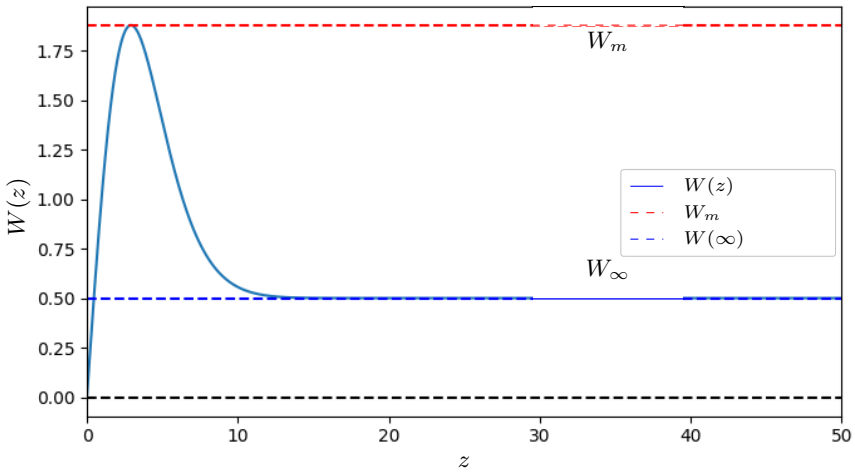


Figure 5.3: **An illustrative integral $W(z)$.** It shows the maximum W_m and the limiting value W_∞. These values depend on the chosen connectivity kernel.

5.4.2 Existence statements and their reasoning

The following conditions describe the existence of three equilibrium types when $P = 0$:

a) A $\varnothing$-solution exists if and only if $h \le 0$.

b) An ∞-solution exists if and only if $2W_\infty + h > 0$.

c) An a-solution exists if and only if $h < 0$ and $a > 0$ satisfies $W(a) + h = 0$.

Let us have a look at how these conditions provide the existence of these three types of equilibrium solutions. We start with the first item. If there exists an $\varnothing$-solution, then (5.4) becomes for the $\varnothing$-solution as follows:

$$u(x,t) = h.$$

This requires that $h \leq 0$ for $u(x,t)$ to be $\varnothing$-solution. So, if there exists an $\varnothing$-solution, then $h \leq 0$ must hold. On the other side, if $h \leq 0$, then $u(x,t) = h$ is indeed an $\varnothing$-solution to (5.4), providing that there exists an $\varnothing$-solution.

We follow a similar reasoning in the second item. If there exists an ∞-solution, then we can write (5.4) for this solution by using the antisymmetry of $W(x)$ as

$$u(x,t) = 2\,W_\infty + h.$$

Since $u(x,t)$ is an ∞-solution, the following holds:

$$2\,W_\infty + h > 0,$$

i.e., $2\,W_\infty > -h$. On the other way around, if $2\,W_\infty > -h$, then $u(x,t) = 2\,W_\infty + h$ is indeed an ∞-solution to (5.4).

In the third item, if there exists a-solution with $R_t(u) = (0,a)$, then (5.4) becomes

$$u(x,t) = \int_0^a s\big(u(y,t)\big)\,\omega(x-y)\,\mathrm{d}y + h = W(x) - W(x-a) + h, \tag{5.5}$$

where we used a change of variables[1] in the integral. Being an a-solution, once it reaches to equilibrium, u satisfies (e.g., as in Figure 5.2c)

$$u(0,t) = u(a,t) = 0, \tag{5.6}$$

This provides $u(a,t) = W(a) - W(0) + h = W(a) + h = 0$. On the other side, if $W(a) + h = 0$, then $u(x,t)$ given by (5.5) satisfies (5.6). Therefore, it is an a-solution.

The graph of W gives a quick way to see which solution types exist for a given h: draw a horizontal line at height $-h$ and read its intersections with $W(z)$ for $z > 0$. This graphical reasoning provides a visual way to classify equilibria as h changes. Figure 5.4 shows three cases. Case I_1 has $2W_\infty > W_m$, case I_2 has $W_m > 2W_\infty > 0$, and case II has $W_\infty < 0$. In the figure, $\varnothing$ denotes a $\varnothing$-solution, ∞ an ∞-solution, a or a_1, a_2 the widths of a-solutions, and the wavy trace represents spatially periodic pattern.

[1] We used the change of variables $\tilde{y} = x - y$, $-\mathrm{d}\tilde{y} = \mathrm{d}y$ in the integral given in (5.4). This provides $\int_0^a \omega(x-y)\,\mathrm{d}y = -\int_x^{x-a} \omega(\tilde{y})\,\mathrm{d}\tilde{y} = \int_{x-a}^{x} \omega(\tilde{y})\,\mathrm{d}\tilde{y} = \int_0^x \omega(\tilde{y})\,\mathrm{d}\tilde{y} - \int_0^{x-a} \omega(\tilde{y})\,\mathrm{d}\tilde{y} = W(x) - W(x-a)$.

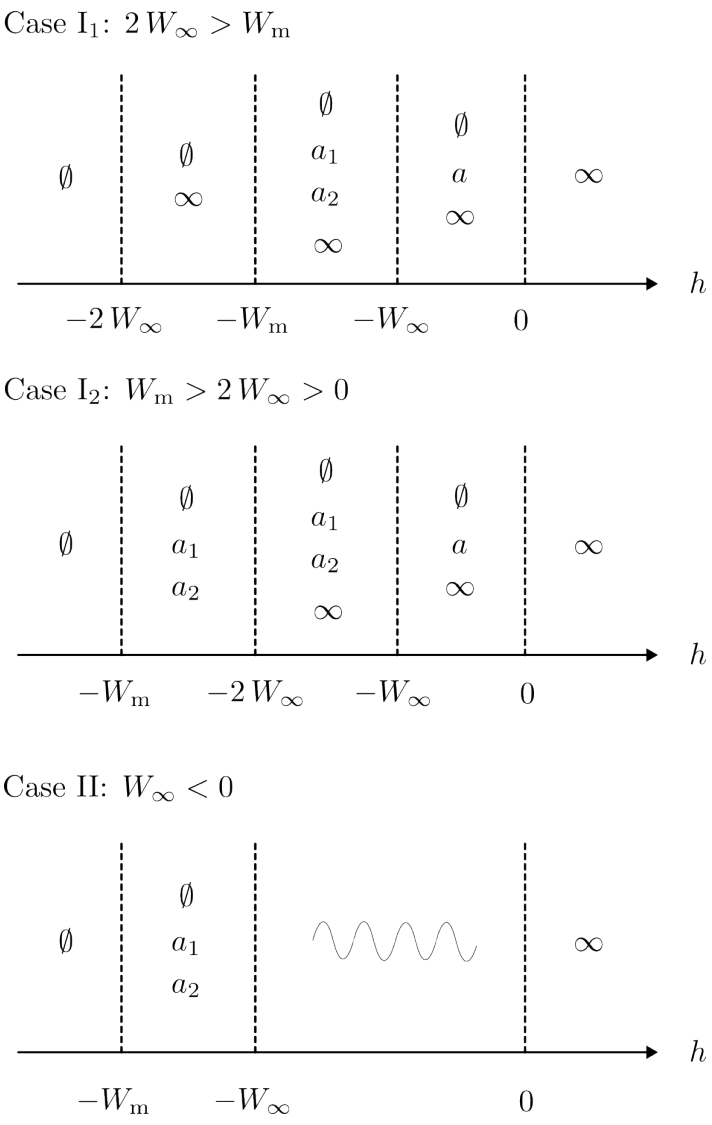


Figure 5.4: **Equilibrium-solution taxonomy as the baseline h varies (adapted from [1]).** The relevant explanations are given in the paragraph preceding the figure.

5.5 Stability of equilibrium solutions

Stability analysis focuses on how a small perturbation in initial condition of a system affects the solution. If a small perturbation in the initial condition does not introduce any change in the equilibrium solution, then the equilibrium solution is stable. Otherwise, it is unstable.

In the following, we will look at the behavior of a single peak of a solution $u(x,t)$, which is not necessarily stationary (not equilibrium). That is, it may be changing with t, so $\dfrac{\partial u(x,t)}{\partial t} = 0$ **need not** necessarily hold. Initial conditions (initial stimulation in Amari's terminology) will be spatially localized stimulation of the system; see Figure 5.5.

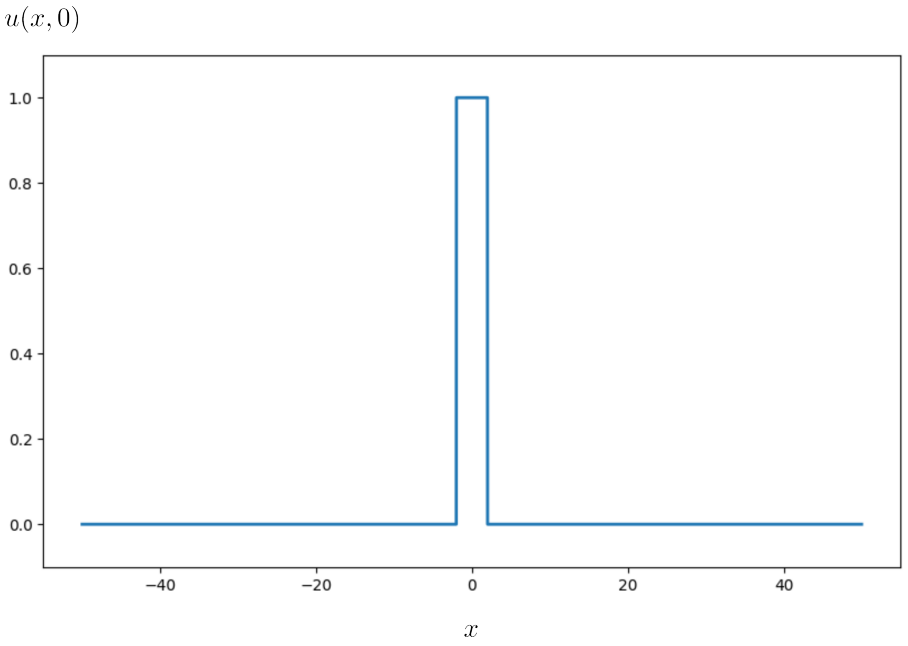


Figure 5.5: **An example of a spatially localized initial stimulus.** It can start a nonstationary response whose active region subsequently changes.

Let the excited region at time $t \geq 0$ be defined as

$$R_t\big(u(x,t)\big) \coloneqq \big(x_1(t), x_2(t)\big).$$

In Figure 5.6, we see the illustration of the spatial gradients at the boundaries of the excited region. These gradients are defined as follows:

$$c_1 \coloneqq \frac{\partial u}{\partial x}(x_1,t), \quad c_2 \coloneqq \frac{\partial u}{\partial x}(x_2,t). \tag{5.7}$$

At the end of a short time interval $\mathrm{d}t$, the solution $u(x,t)$ evolves to $u(x,t+\mathrm{d}t)$; see Figure 5.7. Therefore, the excited region evolves to

$$R_{t+\mathrm{d}t}(u) = \big(x_1(t+\mathrm{d}t), x_2(t+\mathrm{d}t)\big).$$

By definition, a localized excitation should be 0 at the boundaries x_1, x_2 ($x_2 > x_1$) of the excited region:

$$\begin{aligned} &u\big(x_1(t),t\big) = 0, \quad u\big(x_2(t),t\big) = 0, \\ &u\big(x_1(t+\mathrm{d}t),t+\mathrm{d}t\big) = u(x_1+\mathrm{d}x_1,t+\mathrm{d}t) = 0, \\ &u\big(x_2(t+\mathrm{d}t),t+\mathrm{d}t\big) = u(x_2+\mathrm{d}x_2,t+\mathrm{d}t) = 0. \end{aligned} \tag{5.8}$$

Here $\mathrm{d}x_1, \mathrm{d}x_2$ and $\mathrm{d}t$ are infinitesmally small quantities. We approximate $u(x_1+\mathrm{d}x_1, t+\mathrm{d}t)$ and $u(x_2+\mathrm{d}x_2, t+\mathrm{d}t)$ by using their Taylor expansions around (x_1,t) and (x_2,t), respectively:

$$\begin{aligned} u(x_1+\mathrm{d}x_1,t+\mathrm{d}t) &\approx u(x_1,t) + \mathrm{d}x_1 \frac{\partial u}{\partial x}(x_1,t) + \mathrm{d}t \frac{\partial u}{\partial t}(x_1,t), \\ u(x_2+\mathrm{d}x_2,t+\mathrm{d}t) &\approx u(x_2,t) + \mathrm{d}x_2 \frac{\partial u}{\partial x}(x_2,t) + \mathrm{d}t \frac{\partial u}{\partial t}(x_2,t), \end{aligned}$$

from which we can write the following due to (5.8):

$$\begin{aligned} \mathrm{d}x_1 \frac{\partial u}{\partial x}(x_1,t) + \mathrm{d}t \frac{\partial u}{\partial t}(x_1,t) = 0, \\ \mathrm{d}x_2 \frac{\partial u}{\partial x}(x_2,t) + \mathrm{d}t \frac{\partial u}{\partial t}(x_2,t) = 0. \end{aligned} \tag{5.9}$$

Finally, we use (5.8) in (5.1) ($P(x,t) = 0$ for all $x \in (-\infty,\infty)$ and $t \in [0,\infty)$ since we consider the cases in which there is no time-dependent external stimulation) to write

$$\begin{aligned} \mu \frac{\partial u}{\partial t}(x_1,t) &= \int_{x_1}^{x_2} \omega(x_1-y)\,\mathrm{d}y + h = W(x_2-x_1) + h, \\ \mu \frac{\partial u}{\partial t}(x_2,t) &= \int_{x_1}^{x_2} \omega(x_2-y)\,\mathrm{d}y + h = W(x_2-x_1) + h, \end{aligned} \tag{5.10}$$

where we used a change of variables similarly to (5.5), and $W(x) = -W(-x)$, to write the final equalities[2].

From (5.9), we find:

$$\frac{\mathrm{d}x_1}{\mathrm{d}t} = -\frac{\frac{\partial u}{\partial t}(x_1,t)}{\frac{\partial u}{\partial x}(x_1,t)}, \qquad \frac{\mathrm{d}x_2}{\mathrm{d}t} = -\frac{\frac{\partial u}{\partial t}(x_2,t)}{\frac{\partial u}{\partial x}(x_2,t)} \tag{5.11}$$

By using (5.10) and c_1, c_2 defined in (5.7), we write from (5.11) the following:

$$\begin{aligned} \frac{\mathrm{d}x_1}{\mathrm{d}t}(t) = -\frac{\frac{\partial u}{\partial t}(x_1,t)}{\frac{\partial u}{\partial x}(x_1,t)} = -\frac{1}{\mu\, c_1}\Big(W\big(x_2(t)-x_1(t)\big)+h\Big), \\ \frac{\mathrm{d}x_2}{\mathrm{d}t}(t) = -\frac{\frac{\partial u}{\partial t}(x_2,t)}{\frac{\partial u}{\partial x}(x_2,t)} = -\frac{1}{\mu\, c_2}\Big(W\big(x_2(t)-x_1(t)\big)+h\Big). \end{aligned} \tag{5.12}$$

[2]We use the change of variables $\tilde{y} = y - x_1$, which gives $\mathrm{d}\tilde{y} = \mathrm{d}y$. This provides $\int_{x_1}^{x_2} \omega(x_1-y)\,\mathrm{d}y = \int_0^{x_2-x_1} \omega(-\tilde{y})\,\mathrm{d}\tilde{y} = \int_0^{x_2-x_1} \omega(\tilde{y})\,\mathrm{d}\tilde{y}$, where we use the symmetry $\omega(x) = \omega(-x)$. This gives $\int_0^{x_2-x_1} \omega(\tilde{y})\,\mathrm{d}\tilde{y} = W(x_2-x_1)$. The same holds for the second line in (5.10).

Note that the length of the excited region at time t is

$$a(t) = x_2(t) - x_1(t),$$

as shown in Figure 5.7. This implies

$$\frac{\mathrm{d}a}{\mathrm{d}t}(t) = \frac{1}{\mu}\Big(-\frac{1}{c_2} + \frac{1}{c_1}\Big)\Big(W\big(a(t)\big) + h\Big), \tag{5.13}$$

where we used (5.12) and ignored the explicit time notation. This equation describes the dynamical description of the change of the length of the excited region. For an a-solution at equilibrium, the length a does not change in time, therefore $\dfrac{\mathrm{d}a}{\mathrm{d}t} = 0$. This corresponds to an equilibrium length in (5.13). Since $c_1 > 0$ and $c_2 < 0$ for an excited region (see Figures 5.6 and 5.7), this happens only when

$$W(a) + h = 0,$$

which is in agreement with the condition c) given in Section 5.4.2. Moreover, such equilibrium length a is **stable** if $\dfrac{\mathrm{d}W(a)}{\mathrm{d}a} < 0$ and unstable if $\dfrac{\mathrm{d}W(a)}{\mathrm{d}a} > 0$. This allows us to check the stability of equilibrium solutions easily by using the plot of $W(x)$; see Figure 5.3.

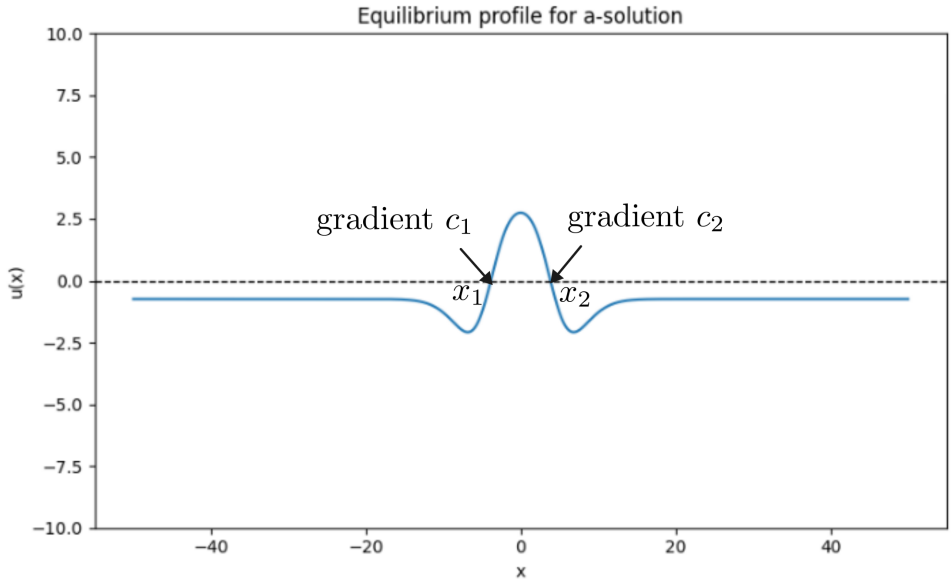


Figure 5.6: **Spatial gradients c_1 (positive) and c_2 (negative) at the boundaries of the localized excitation.**

Example 1. *We can think of a case where h is chosen such that there are two a values a_1 and a_2 $(a_2 > a_1)$ for an a-solution:*

$$W(a_1) + h = 0, \qquad W(a_2) + h = 0.$$

Inspection of the stability of these solution can be done directly on the corresponding $W(x)$ graph. In Figure 5.8, we see that the gradient for W at a_1 is positive ($\frac{\mathrm{d}W}{\mathrm{d}a}(a_1) > 0$). Therefore, a_1 is an unstable equilibrium of (5.13), i.e., a_1-solution is an unstable equilibrium solution. This is the opposite for a_2, therefore a_2 is a stable equilibrium of (5.13), thus a_2-solution is a stable equilibrium solution.

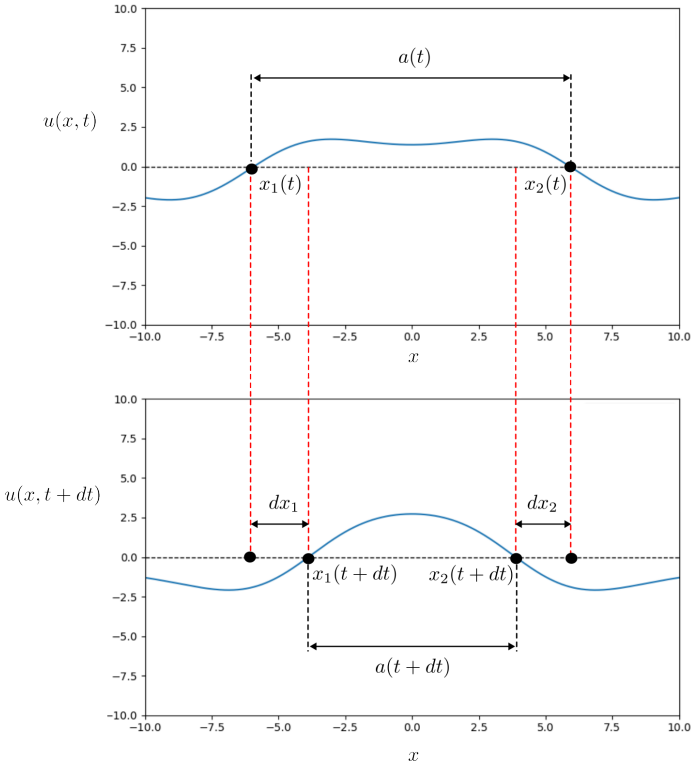


Figure 5.7: **Evolution of a localized excitation over a short time interval** $\mathrm{d}t$ **before reaching equilibrium.** The positions of the excitation boundaries at times t and $t+\mathrm{d}t$ are denoted by $x_1(t)$, $x_2(t)$ and $x_1(t+\mathrm{d}t)$, $x_2(t+\mathrm{d}t)$, respectively. The displacements of these boundaries over the time interval $\mathrm{d}t$ are denoted by $\mathrm{d}x_1$ and $\mathrm{d}x_2$, respectively. The lengths of the excited region at times t and $t+\mathrm{d}t$ are given by $a(t)$ and $a(t+\mathrm{d}t)$, respectively.

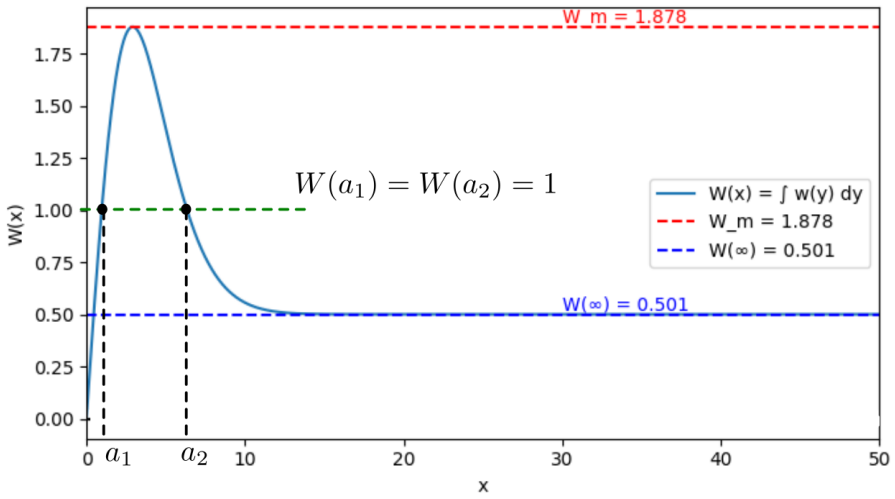


Figure 5.8: **Plot of $W(x)$ where h is chosen ($= -1$) such that $W(a_1) + h = W(a_2) + h = 0$.** This plot provides a graphical explanation of Example 1.

For a finite excited region to explode towards an ∞-solution, the following condition is necessary:

$$W_\infty + h > 0. \tag{5.14}$$

If this condition is not satisfied, an ∞-solution cannot be obtained without stimulating infinitely large regions.

Therefore, we ignore ∞-solution in the cases where (5.14) is not satisfied. This simplifies the taxonomy given in Figure 5.4, removing the elements with $h < -W_\infty$ from Cases I_1 and I_2. So the Cases I_1 and I_2 overlap. We provide in Figure 5.9 this simplified version in which we denote the unstable equilibrium solutions with the hat sign $\hat{}$.

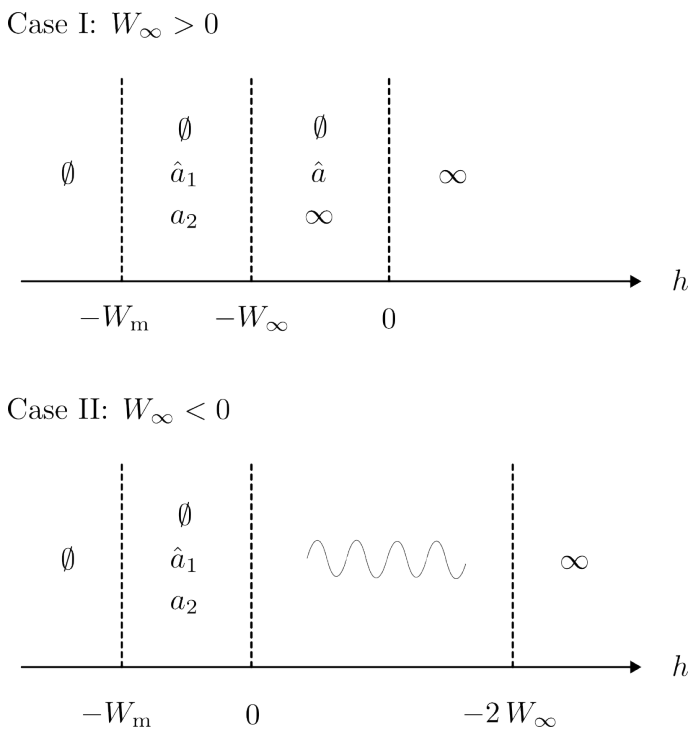


Figure 5.9: **The simplified taxonomy of equilibrium solutions and their stability (adapted from [1]).** The hat sign ˆ indicates the unstable solutions.

5.6 Scope of the model

The Amari model keeps spatially distributed neural interactions while reducing the state variables to one average membrane-potential variable. Its Heaviside version lets us locate active regions and analyze the width of a persistent a-solution with the simple function W.

Chapter 6

Neurogeometry

The previous chapters described neural activity as a function of time and, later, of position. In primary visual cortex (V1), position is not the only useful variable. Many neurons respond selectively to the *orientation* of a line or edge. Neurogeometry studies how visual features such as position and orientation can be represented together and how that representation relates to the functional organization of V1. In this chapter, we consider a static, orientation-based geometric construction. This construction provides the framework for a model of an early stage of visual processing found in many species, such as cats and primates.

6.1 From the retina to orientation-selective cells

The visual pathway starts in the retina. Light is detected in the retina. Retinal cells generate signals in response to light projected onto the retinal layer. These signals travel through the optic nerve and optic chiasm. The optic chiasm helps organize information from the two eyes. The signals then continue along the optic tract to the lateral geniculate nucleus (LGN) of the thalamus. The LGN processes and relays the signals coming from the optic chiasm. The signals finally reach V1 through the optic radiation, see Figure 6.1. In V1, they stimulate the cortical neurons which are sensitive to certain visual features such as orientation.

V1 lies in the occipital lobe of the brain. Its neurons begin to extract features from the visual signal. This was demonstrated in experimental recordings of visual cortical neurons which were presented by David H. Hubel and Torsten Wiesel in their classical work [8]. A particularly important feature is the orientation of a local contrast boundary.

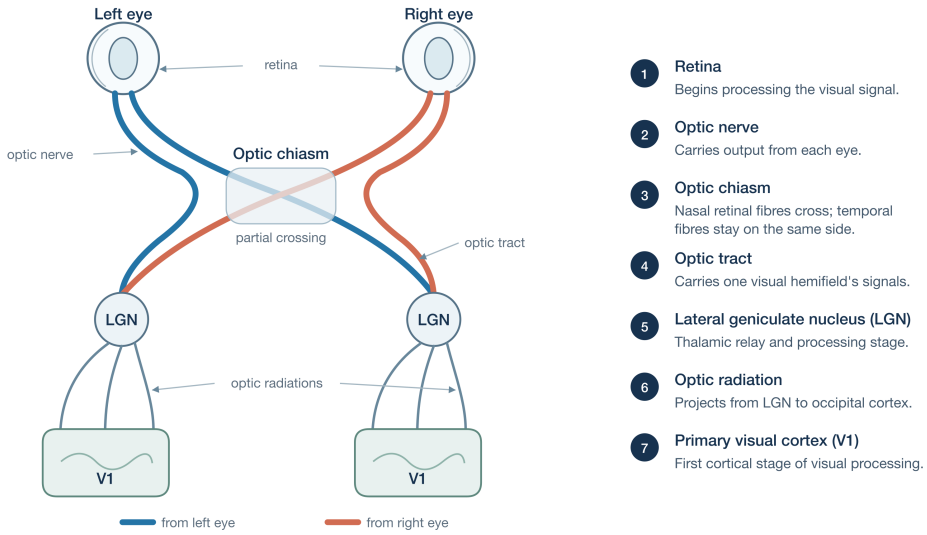


Figure 6.1: **Bilateral visual pathway from the eyes to the primary visual cortex (V1).**

6.2 Receptive fields, profiles, and orientation selectivity

The *receptive field* of a visual neuron is the region of the input image where a stimulus can affect that neuron. A *receptive profile* gives more information: it describes how different positions within that region contribute positively or negatively to the response. In the simplified picture of Figure 6.2, an LGN neuron has a roughly circular center-surround organization. A V1 orientation-sensitive neuron (*simple cell*) has elongated subregions. In a 2D image, a bright bar that is placed along a positive subregion of the simple cell can produce a stronger response than the same bar placed across it. Thus a simple cell can be selective for orientation as well as position.

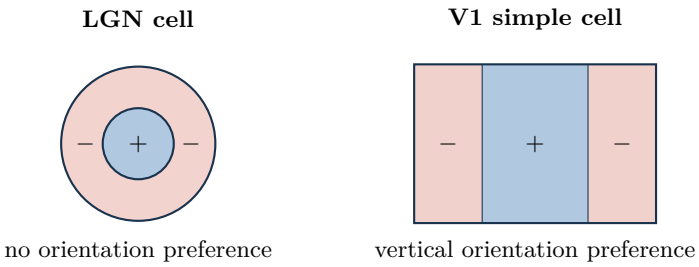


Figure 6.2: **Schematic receptive profiles.** Blue + and red − indicate regions with opposite effects for a visual stimulus. The arrangement of the subregions of the simple cell receptive profile is the main ingredient for the orientation-selectivity of the simple cells .

Orientation selectivity is graded, not an all-or-none label: a neuron generally responds most strongly near its preferred orientation and less strongly as the stimulus is rotated away from the preferred orientation. The response also depends on contrast and the position of the bar within the receptive field. The simple mathematical model below isolates orientation and position so

that their roles are easy to see.

6.3 Gabor function models a simple-cell receptive profile

Several families of filters can approximate simple-cell receptive profiles, including Gaussian derivatives, cake kernels, and in particular Gabor functions. We use Gabor functions because they combine a localizing Gaussian window with an oscillation that creates alternating positive and negative subregions. They are also convenient because a whole family of profiles can be generated by translating and rotating one reference profile.

Let $p = (x_0, y_0)$ be the center of a receptive profile and let $\theta \in [0, \pi)$ be its preferred orientation. For an image point at $q = (x, y)$, define coordinates centered at p and rotated with the profile:

$$\begin{pmatrix} \xi \\ \eta \end{pmatrix} = R_{-\theta} \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix}, \qquad R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}. \tag{6.1}$$

Thus $\xi = (x - x_0)\cos\theta + (y - y_0)\sin\theta$ measures a displacement along the axis aligned with the preferred orientation, while $\eta = -(x - x_0)\sin\theta + (y - y_0)\cos\theta$ measures the displacement across it. A Gabor profile is given by

$$\Psi_{p,\theta}(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{\xi^2 + \eta^2}{\sigma^2}\right) \exp\left(\mathrm{i}\frac{\eta}{\sigma}\right), \qquad \sigma > 0. \tag{6.2}$$

The Gaussian component restricts the profile to a neighborhood of p, therefore making the receptive profile localized around its center (x_0, y_0). The complex exponential supplies a wave across the axis aligned with the preferred orientation. Increasing σ increases the scale of the Gaussian, causing the profile to spread over a larger region and thus become less localized.

Using $\exp(\mathrm{i}z) = \cos z + \mathrm{i}\sin z$, we can see that the real part of (6.2) is an even cosine-like profile across η, and the imaginary part is an odd sine-like profile. Figure 6.3 shows one-dimensional sections through these two components. In two dimensions, the alternating bands extend along ξ. A bright bar aligned with those bands has a different overlap with the filter from a bar crossing them.

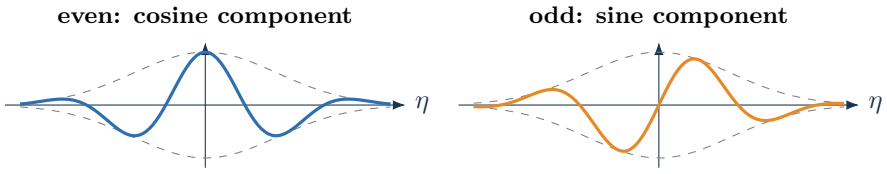


Figure 6.3: **Cross-sections of the even and odd parts of a Gaussian-windowed oscillation.** Dashed curves show the Gaussian envelope. The horizontal coordinate η is perpendicular to the preferred orientation of the receptive profile.

Write $\Psi_{\mathbf{0},0}$ for the reference, or *mother*, Gabor profile centered at the origin with zero rotation. Equation (6.1) gives

$$\Psi_{p,\theta}(q) = \Psi_{\mathbf{0},0}\big(R_{-\theta}(q - p)\big).$$

Once this transformation is applied, the same kind of local measurement can be made at every image position and orientation by using the reference profile $\Psi_{\mathbf{0},0}$. Since an orientation has no direction, the orientation variable is taken modulo π.

Finally, it is important to note that using the Gabor functions as mathematical representations of receptive profiles is a modeling choice. This is not a claim that a biological simple cell exactly implements the Gabor function given by (6.2).

6.4 Orientation columns and orientation fibres

Each Gabor function $\Psi_{p,\theta}$ has a position parameter with two components, and one orientation parameter. The visual cortex also contains neurons with different preferred orientations at nearby cortical locations. In recordings across V1, orientation preference gradually changes as an electrode moves across the cortical surface, while neurons encountered along a perpendicular movement of the electrode to the surface can have similar orientation preferences. This motivates the idea of orientation columns, see Figure 6.4.

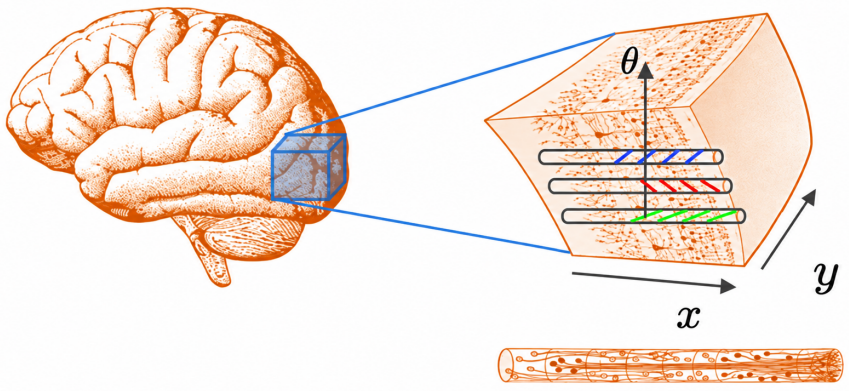


Figure 6.4: **Orientation-selective organization in V1.** The enlarged cortical region illustrates orientation columns and the position-orientation coordinates (x, y, θ) used in the model.

We restrict ourselves to 2D gray scale images as visual stimuli. For a fixed visual position p on the 2D image plane, we consider a collection of orientation-selective channels indexed by θ. This collection is called the *orientation fibre* over p. The position-orientation space is described by

$$\mathcal{N} = M \times [0, \pi), \qquad p = (x, y) \in M,$$

with 0 and π identified as the same orientation due to the π modulo structure. Here $M \subset \mathbb{R}^2$ is the 2D image plane. It is assumed to be identical to the retinal surface and to the V1 surface. Every point of M has its own fibre, so the model can ask both *where* a feature is and *which way* it is oriented. Figure 6.5 illustrates this idea.

The neurogeometric approach uses this position-orientation representation as a starting point for geometric models of visual organization and perceptual completion [4, 10, 11]. This repre-

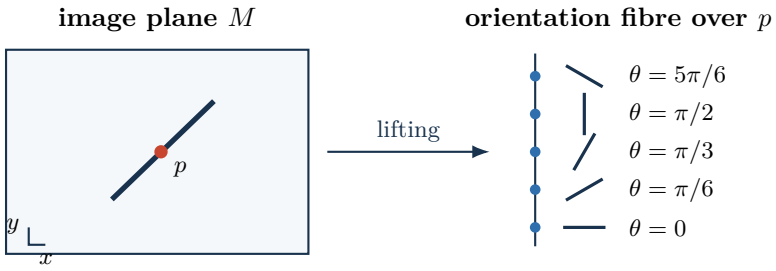


Figure 6.5: **At one image position p, the model keeps a set of channels with different preferred orientations.** The five displayed angles are samples of a continuous interval $[0, \pi)$.

sentation is closely related to an operation called *lifting*, see Figure 6.6. This operation models the output responses of the simple cells in orientation columns once they are stimulated by a visual stimulus.

6.5 Lifting an image into position-orientation space

To simplify the biological picture for a manageable model, we make the following assumptions:

1. The input is a single, two-dimensional grayscale image $I : M \to R$. Time, motion in the scene, and eye or head movement are omitted.
2. We identify image, retinal, and cortical position coordinates with the same plane M. In reality, the retino-cortical map changes coordinates. The identification by using the same plane for the image, retinal and cortical surfaces is for modeling convenience, it does not closely match with the anatomy.
3. Receptive profiles are static, they do not change in time. Moreover, they take into account only orientation selectivity, leaving out other important feature selectivity such as spatial frequency and scale selectivity.
4. The initial response is a linear filter output. Real neurons also have thresholds, saturation, adaptation, and spiking dynamics, none of which are described in the model framework.

These assumptions separate the geometry in which the visual features are represented from the dynamics of a neural population. In particular, the model below tells us how to compute an orientation-indexed response from an image; it does not yet specify how V1 responses interact or evolve in time.

Let $I(x, y)$ be a grayscale image. For each $p = (x_0, y_0)$ and $\theta \in [0, \pi)$, filter the image with the corresponding receptive profile modeled by (6.2). The neural response $O : M \times [0, \pi) \to C$ to I is obtained as the output of the linear filtering based on

$$O(p, \theta) = \int_M I(x, y)\, \Psi_{p,\theta}(x, y)\, \mathrm{d}x\, \mathrm{d}y. \tag{6.3}$$

This formula is a convolution. Image intensity values at positions where the filter is positive and negative contribute with opposite signs. Computing (6.3) for all p and θ turns a 2-dimensional

input into a function on the 3-dimensional space $\mathcal{N}$. This is called *lifting*. At one position, the values $O(p, \theta)$ form an orientation-tuning profile. They form an orientation-indexed representation, see Figure 6.6.

Since the Gabor function in (6.2) is complex, O is complex too. We choose either its real or imaginary component in the model framework. This choice depends on the modeling question.

One can depict a return from the lifted representation to the image plane by combining the orientation channels. It is useful to visualize this as collapsing the orientation fibre at every p. However, the expression $\int_0^\pi O(p, \theta)\, \mathrm{d}\theta$ is not, in general, an exact reconstruction of $I(p)$ from the lifted representation. Exact reconstruction requires an appropriate filter family with different frequencies and normalization.

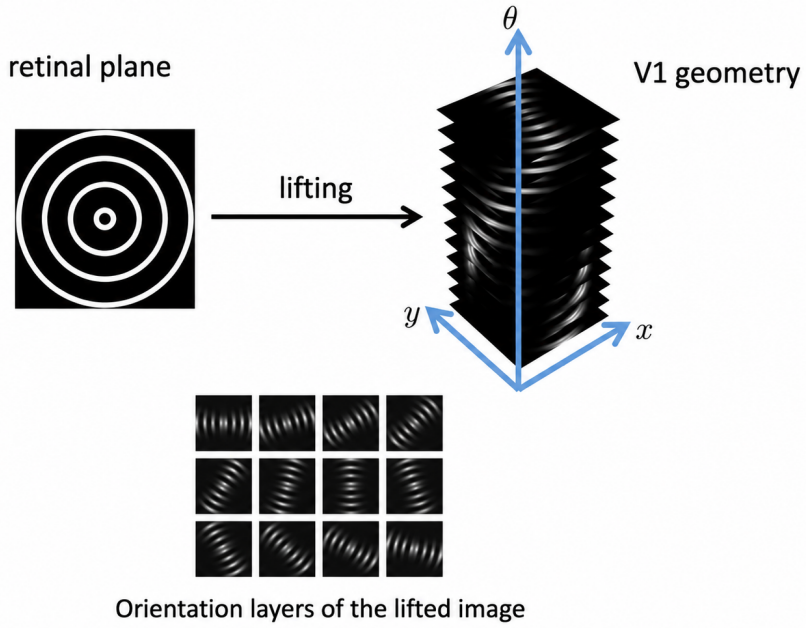


Figure 6.6: **Lifting a retinal image into position-orientation space (adapted from [2]).** Filtering the image at different preferred orientations θ produces the orientation-indexed response $O(x, y, \theta)$. The panels below the schematic show orientation layers of the lifted image.

Bibliography

[1] S. Amari. Dynamics of pattern formation in lateral-inhibition type neural fields. *Biological Cybernetics*, 27(2):77–87, 1977.

[2] E. Baspinar. *Minimal surfaces in sub-Riemannian structures and functional geometry of the visual cortex*. Phd Thesis, Alma Mater Studiorum - Universitá di Bologna, 2018.

[3] N. Brunel and V. Hakim. Fast global oscillations in networks of integrate-and-fire neurons with low firing rates. *Neural Computation*, 11(7):1621–1671, 1999.

[4] G. Citti and A. Sarti. A cortical based model of perceptual completion in the roto-translation space. *Journal of Mathematical Imaging and Vision*, 24(3):307–326, 2006.

[5] M. di Volo, A. Romagnoni, C. Capone, and A. Destexhe. Biologically realistic mean-field models of conductance-based networks of spiking neurons with adaptation. *Neural Computation*, 31(4):653–680, 2019.

[6] J. Eccles. The physiology of synapses. *Berlin, Göttingen, Heilderberg*, 1964.

[7] B. J. Fischer, C. H. Anderson, and J. L. Peña. Multiplicative auditory spatial receptive fields created by a hierarchy of population codes. *PLOS One*, 4(11):e8015, 2009.

[8] D. H. Hubel and T. N. Wiesel. Receptive fields of single neurones in the cat's striate cortex. *The Journal of Physiology*, 148(3):574–591, 1959.

[9] B. H. Jansen and V. G. Rit. Electroencephalogram and visual evoked potential generation in a mathematical model of coupled cortical columns. *Biological Cybernetics*, 73(4):357–366, 1995.

[10] J. Petitot. *Elements of Neurogeometry: Functional Architectures of Vision*. Springer International Publishing, 2017.

[11] J. Petitot and Y. Tondut. Vers une neurogéométrie. Fibrations corticales, structures de contact et contours subjectifs modaux. *Mathématiques, Informatique et Sciences Humaines*, 145:5–101, 1999.

[12] H. R. Wilson and J. D. Cowan. Excitatory and inhibitory interactions in localized populations of model neurons. *Biophysical Journal*, 12(1):1–24, 1972.

[13] H. R. Wilson and J. D. Cowan. A mathematical theory of the functional dynamics of cortical and thalamic nervous tissue. *Kybernetik*, 13(2):55–80, 1973.